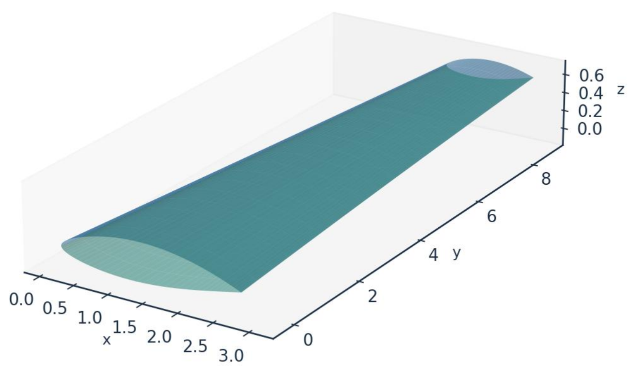
INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, M=0.42, Re_MAC ≈ 6.4×10⁶):

Quantity	Predicted value
Upper-surface transition x_tr/c	0.43 (natural / Tollmien–Schlichting)
Lower-surface transition x_tr/c	0.58 (natural / Tollmien–Schlichting)
Mean laminar-flow extent	≈ 52 % of chord
Section lift coefficient C_l	0.502
Section profile drag C_d	27.0 counts
Viscous drag reduction vs fully-turbulent	≈ 45 %
Climb (Tu=0.9 %) transition x_tr/c (upper)	0.05 (bypass)

Table 1. Headline predictions.

Validated against three independent, credible published transition datasets with one universal calibration set, the solver reproduces the transition-onset Reynolds number Re_θt against the Rolls-Royce hot-wire measurements of ERCOFTAC case 020 and the Schubauer-Skramstad experiment. The natural and bypass branches are independent models; the separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, governed by the growth of the amplification envelope and predicted here by an e^N integral in Re_theta.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f, momentum thickness θ, shape factor H, intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e, with a Prandtl-Glauert correction applied to C_p and the integrated loads at the cruise Mach number. The boundary-layer closures downstream are incompressible formulations and the edge velocity passed to them is left uncorrected.
- Laminar boundary layer: Thwaites' integral method advances θ, H and Re_θ from the stagnation point along each surface.
- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ.
- Turbulent boundary layer: Head's entrainment method with the Ludwieg–Tillmann skin-friction law advances the turbulent BL to the trailing edge.

- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethe & Chow)

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^{N} C_{t,ij}\gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left(\frac{V}{U_\infty}\right)^2$$

[missing eq E04]

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6}\int_0^x U_e^5\,dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu}\frac{dU_e}{dx}, Re_\theta = \frac{U_e\theta}{\nu}$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2+H)\frac{\theta}{U_e}\frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the local Tu; natural/TS onset integrates the envelope amplification rate of Drela & Giles in Re_theta and triggers at N_crit; envelope; separation-induced and cross-flow onsets use bubble and C1 criteria. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (low-Tu correlation surrogate)

$$N(x) = \int_{x_0}^{x} -\alpha_i(x')dx' \geq N_{crit}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin\Lambda \cos\Lambda \ \geq \ C_1$$

(E12) Separation-induced onset (laminar bubble)

$$\lambda \leq -0.09 \Rightarrow Re_{\theta t,sep} = \max\left(Re_{\theta t}^{floor}, 0.7 Re_{\theta t,bp}\right)$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^{*} = \min\left(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF}\right)$$

(E14) Transition trigger

$$Re_\theta(x_t) \geq Re_{\theta t}^{*}$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1-\gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e}\frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwieg-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2\frac{\theta_{TE}}{c}\left(\frac{U_{e,TE}}{U_\infty}\right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r\frac{\gamma - 1}{2} M_e^2 \left[1 - \left(\frac{u}{U_e}\right)^2\right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e\left(1 + r\frac{\gamma - 1}{2}M_e^2\right), r \approx Pr^{1/3}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3}c_{root}\frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e^N codes	RANS γ–Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (envelope e^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble criterion)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. The separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here — the novelty claim.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mach	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (FL100, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input).

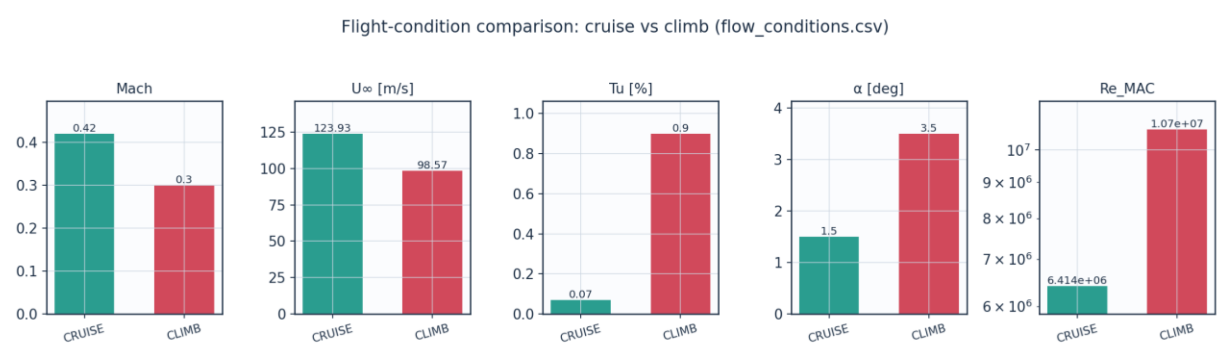


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K

Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) | bypass(AGS@Tu) | separation | crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Separation criterion	Thwaites lambda<=-0.09 / H>2.6 turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	validated on Schubauer-Skramstad (NACA Rep.909)
A_BP	1.0	bypass onset weight	validated on ERCOFTAC T3A/T3B
A_SEP	1.0	separation-induced weight	Mayle (1991) bubble correlation
A_CF	1.0	crossflow weight	Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack correlation N=-8.43-2.4 ln(Tu); 9.00 at Tu=0.07%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al. C1 criterion
CF_ratio	0.356	theta2/theta surrogate for crossflow	calibrated, not validated
sep_floor	120.0	min separation onset Re_theta	calibration floor

Table 7. Calibration constants and their sources (input).

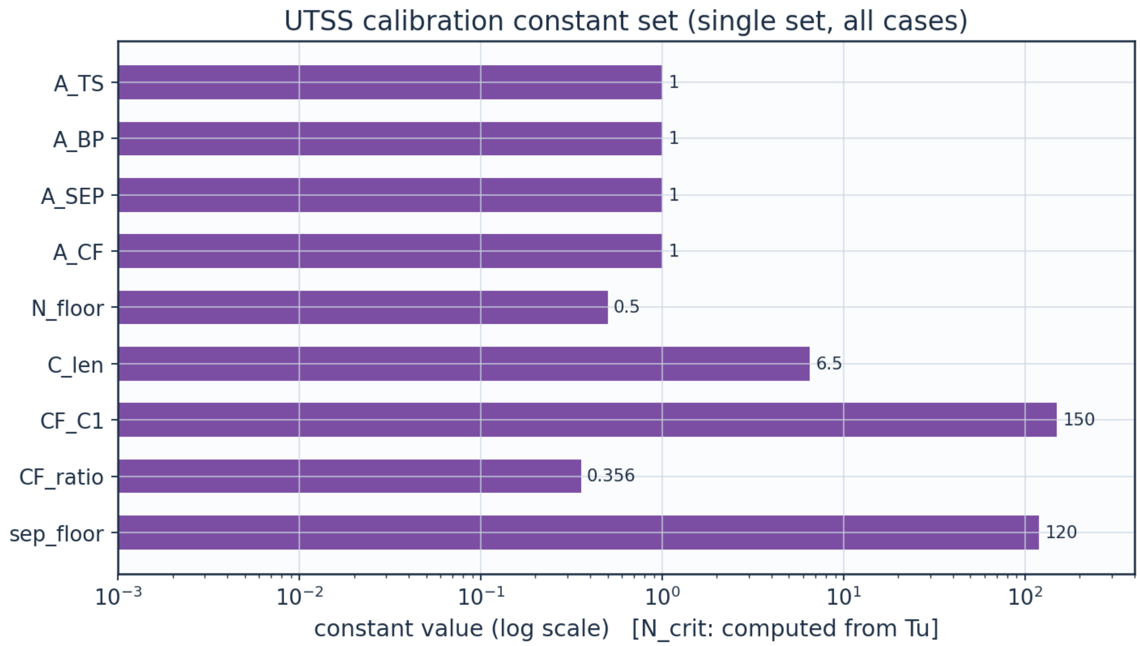


Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

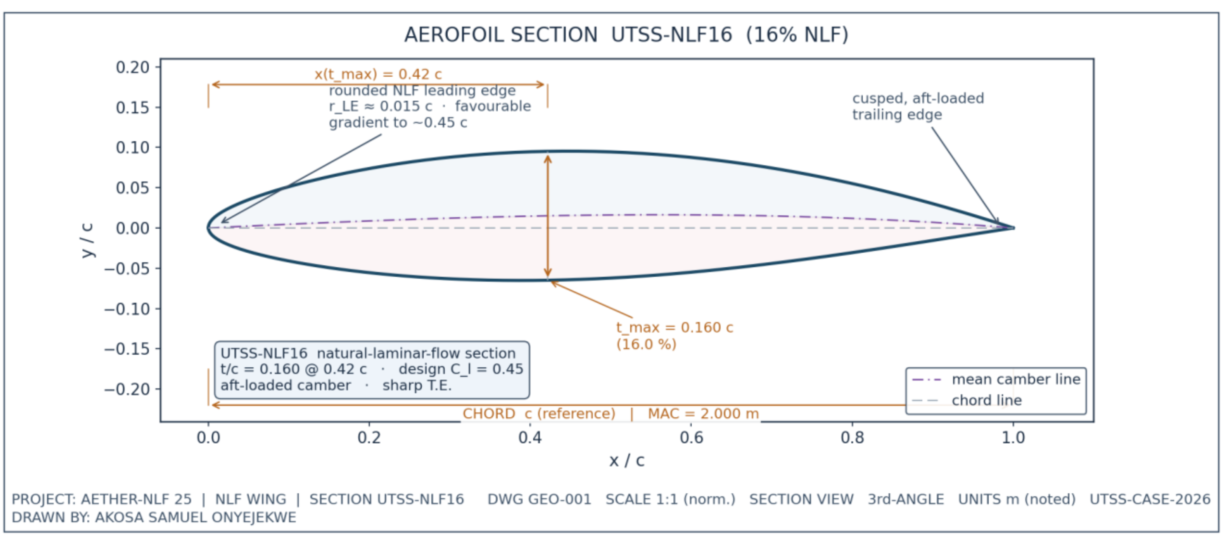


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

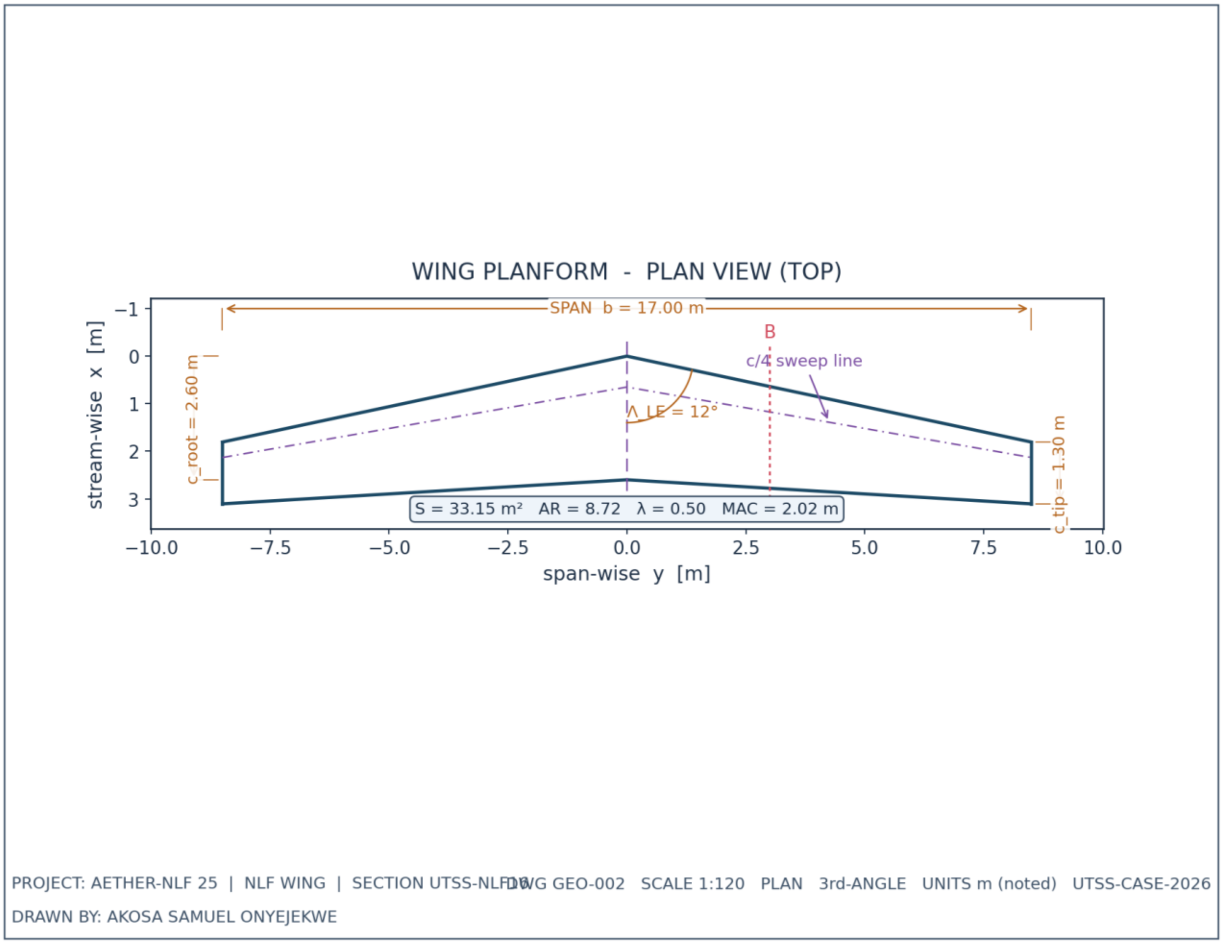


Fig. 2. Wing planform — plan (top) view, dimensioned.

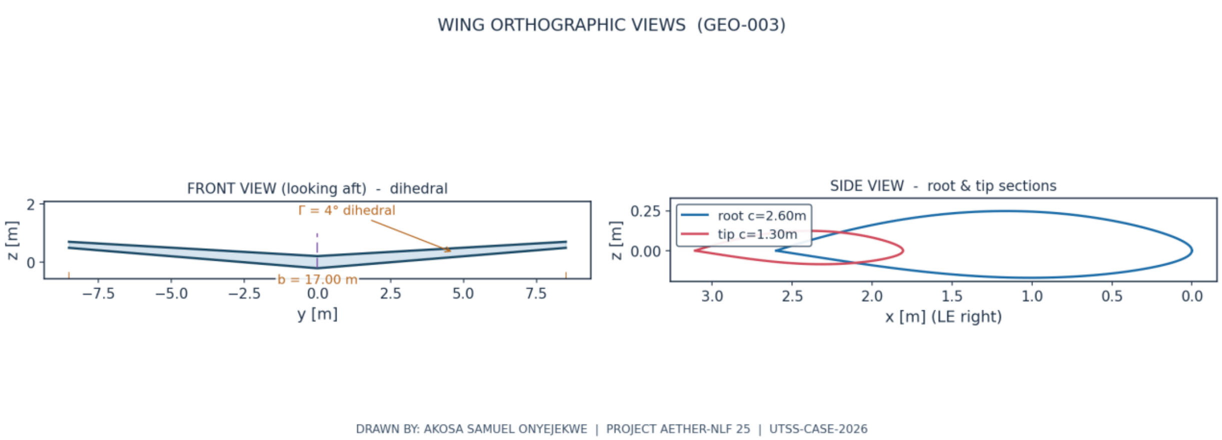


Fig. 3. Front and side orthographic views.

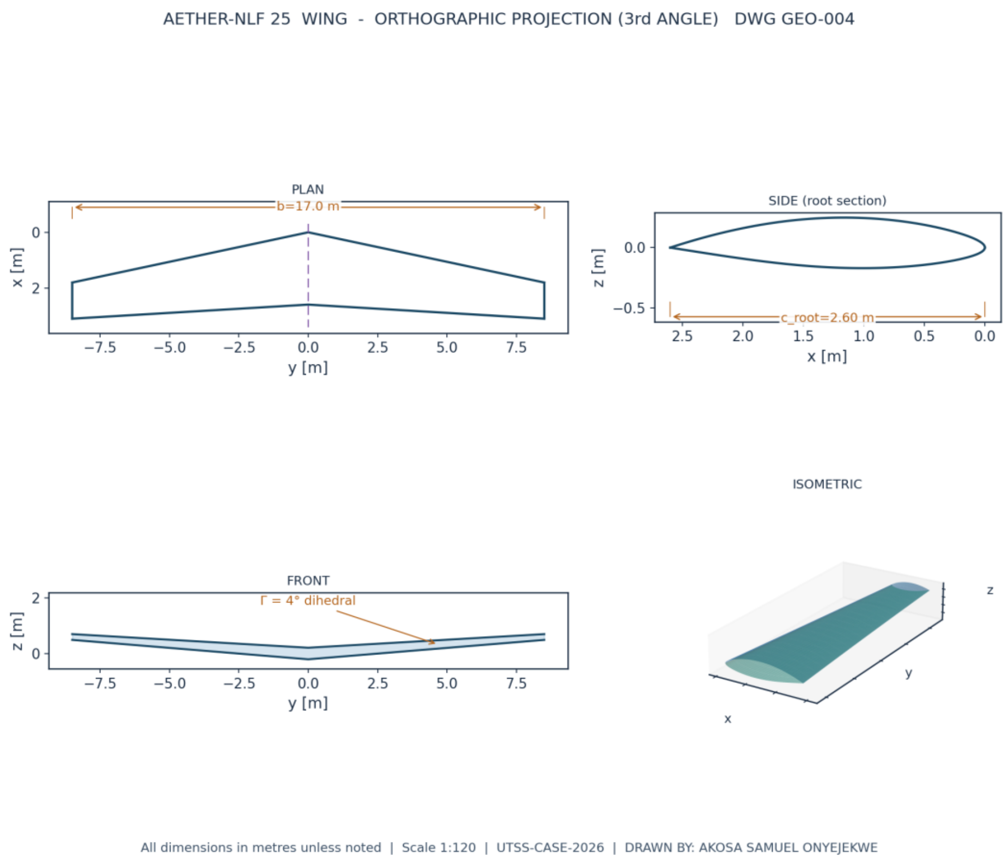


Fig. 4. Full third-angle orthographic projection sheet.

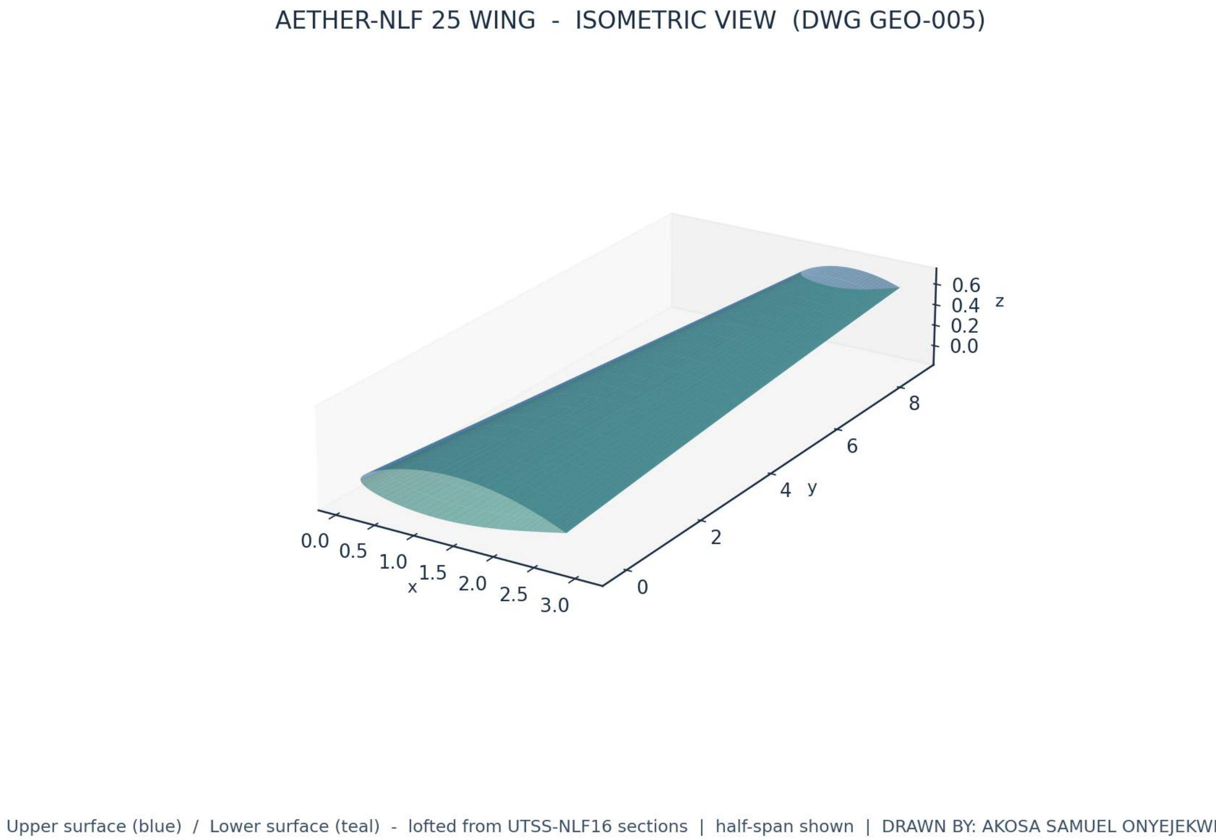


Fig. 5. Isometric view of the wing.

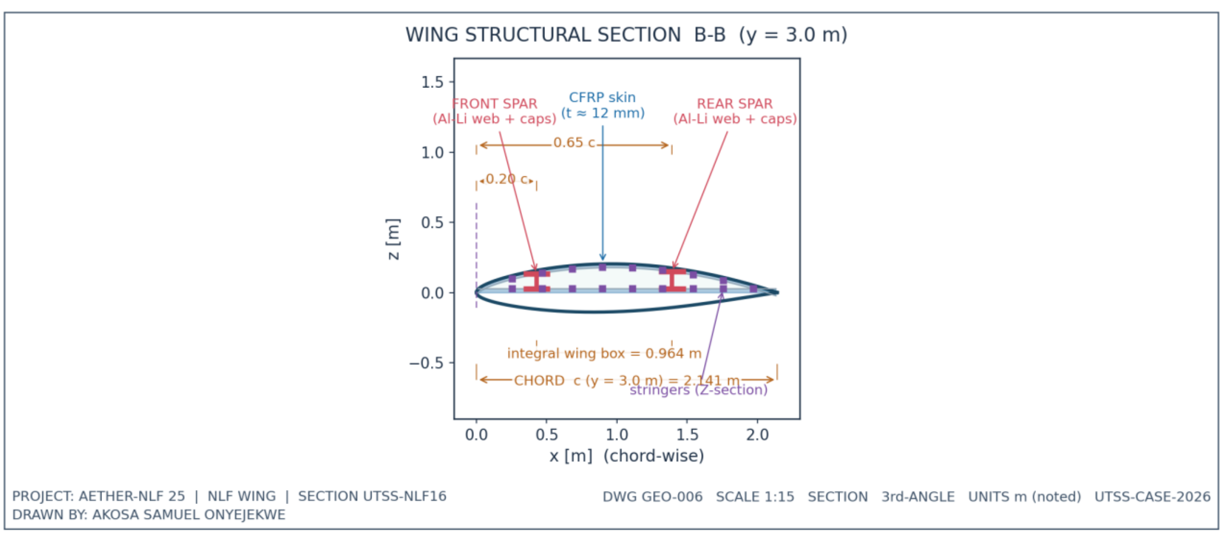


Fig. 6. Structural sectional view B–B.

7.1 Geometry data (from CSV)

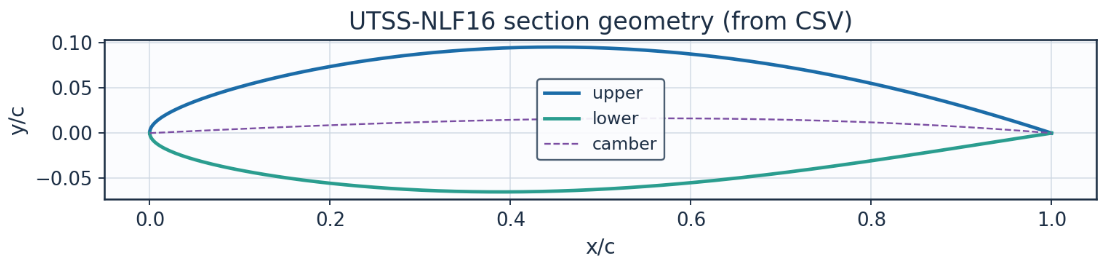


Fig. 7. Section geometry plotted from airfoil_UTSS-NLF16.csv.

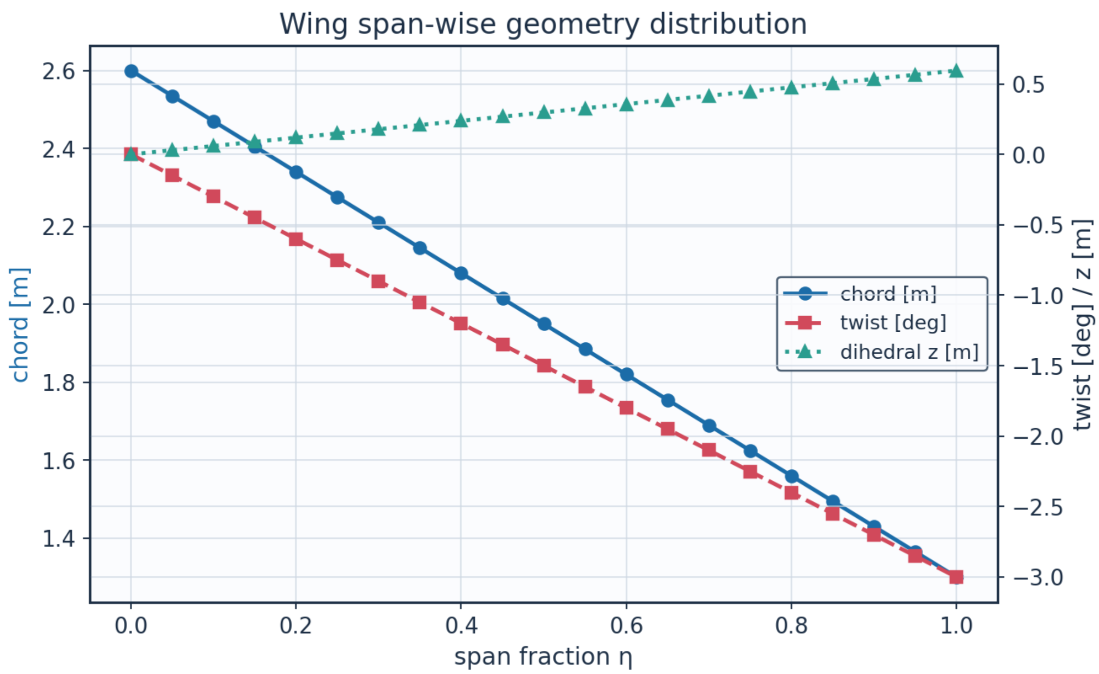


Fig. 8. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m	Re_local
0.0	0.0	2.6	0.0	2.6	-0.0	0.0	8246203.012995781
0.05	0.425	2.535	0.0903365387097594	2.62533653870976	-0.15	0.0297188950759919	8040047.9376708865
0.1	0.8500000000000001	2.47	0.1806730774195188	2.650673077419519	-0.3	0.0594377901519838	7833892.862345993
0.15	1.275	2.405	0.2710096161292782	2.6760096161292783	-0.45	0.0891566852279757	7627737.787021098
0.2	1.7000000000000002	2.34	0.3613461548390376	2.7013461548390376	-0.6000000000000001	0.1188755803039677	7421582.711696202
0.25	2.125	2.275	0.451682693548797	2.726682693548797	-0.75	0.1485944753799596	7215427.636371308
0.3	2.5500000000000003	2.21	0.5420192322585565	2.7520192322585566	-0.9000000000000001	0.1783133704559515	7009272.561046414

0.35	2.975	2.145	0.6323557709683159	2.777355770968316	-1.05	0.2080322655319434	6803117.485721519
0.4	3.4000000000000004	2.08	0.7226923096780753	2.802692309678076	-1.2000000000000002	0.2377511606079354	6596962.410396625
0.45	3.825	2.015	0.8130288483878346	2.828028848387835	-1.35	0.2674700556839273	6390807.33507173
0.5	4.25	1.95	0.903365387097594	2.853365387097594	-1.5	0.2971889507599192	6184652.2597468365
0.55	4.675000000000001	1.885	0.9937019258073536	2.878701925807354	-1.65	0.3269078458359112	5978497.184421942
0.6000000000000001	5.1000000000000005	1.82	1.084038464517113	2.9040384645171127	-1.8000000000000005	0.3566267409119031	5772342.109097046
0.65	5.525	1.755	1.1743750032268725	2.9293750032268724	-1.95	0.386345635987895	5566187.033772152
0.7000000000000001	5.95	1.69	1.2647115419366315	2.954711541936632	-2.1	0.4160645310638869	5360031.958447257
0.75	6.375	1.625	1.3550480806463912	2.980048080646391	-2.25	0.4457834261398788	5153876.883122363
0.8	6.800000000000001	1.56	1.4453846193561506	3.00538461935615	-2.4000000000000004	0.4755023212158709	4947721.807797468
0.8500000000000001	7.225	1.495	1.53572115806591	3.03072115806591	-2.5500000000000003	0.5052212162918628	4741566.732472573
0.9	7.65	1.43	1.6260576967756692	3.056057696775669	-2.7	0.5349401113678547	4535411.65714768
0.95	8.075000000000001	1.365	1.716394235485429	3.081394235485429	-2.85	0.5646590064438467	4329256.581822785
1.0	8.5	1.3	1.806730774195188	3.106730774195188	-3.0	0.5943779015198385	4123101.50649789

Table 8. Wing planform stations (wing_planform.csv).

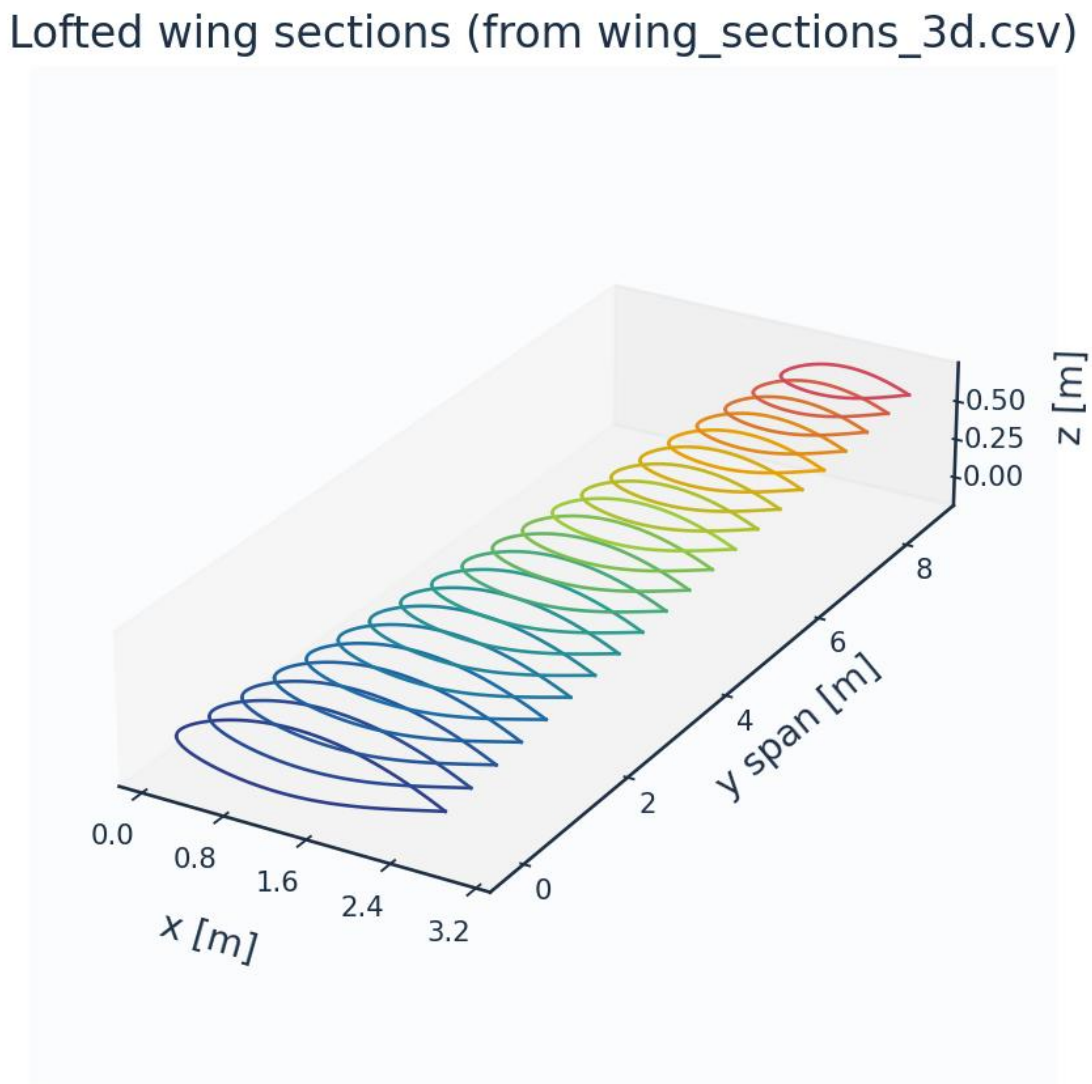


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^{+}\approx1$) supports boundary-layer profile recovery. A mesh-independence study confirms convergence.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)

LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.4996087807380917	0.0025687913527442	0.4334642453558481	
180	0.5010076226503911	0.0026006899204084	0.4378179657089841	1.241773397835555
260	0.5018338859371162	0.002695941579585	0.4324359765532054	3.662553479718112
360	0.5023435792157286	0.0027647311544644	0.4334408286728071	2.5515973862447083
480	0.5026750206350319	0.0028348489572751	0.4301662322757815	2.5361526634316567

Table 10. Mesh-independence study.

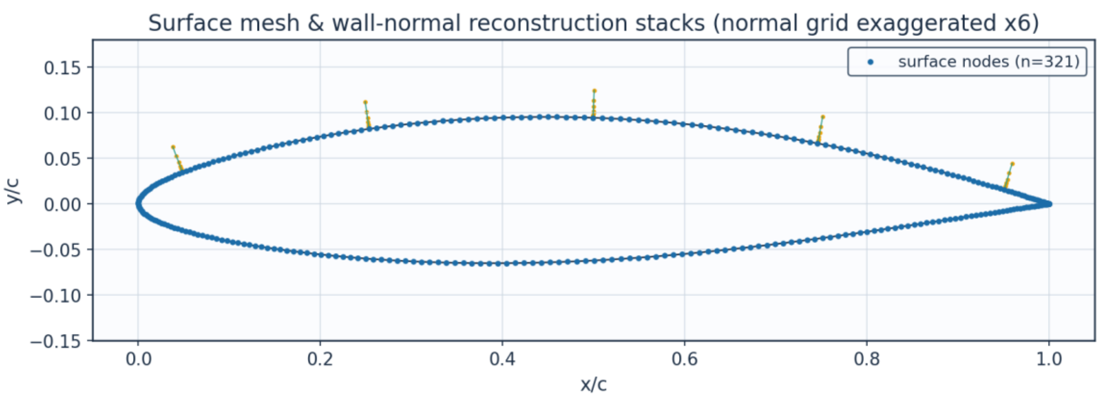


Fig. 9. Surface mesh and wall-normal stacks.

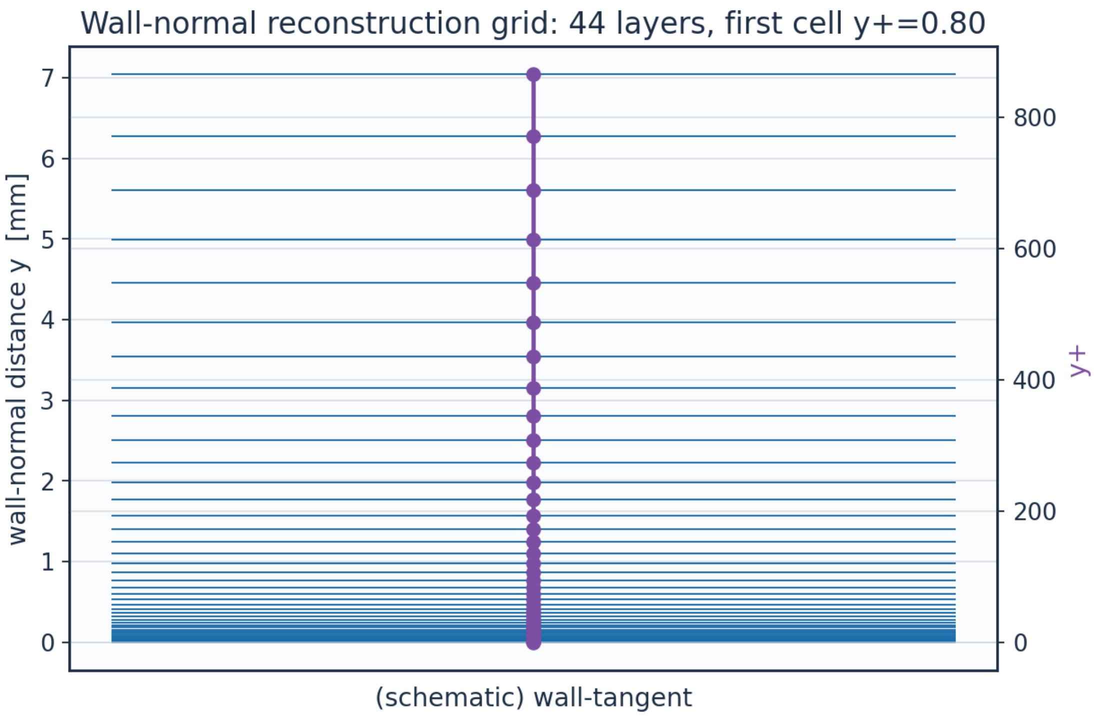


Fig. 10. Wall-normal reconstruction grid / y⁺.

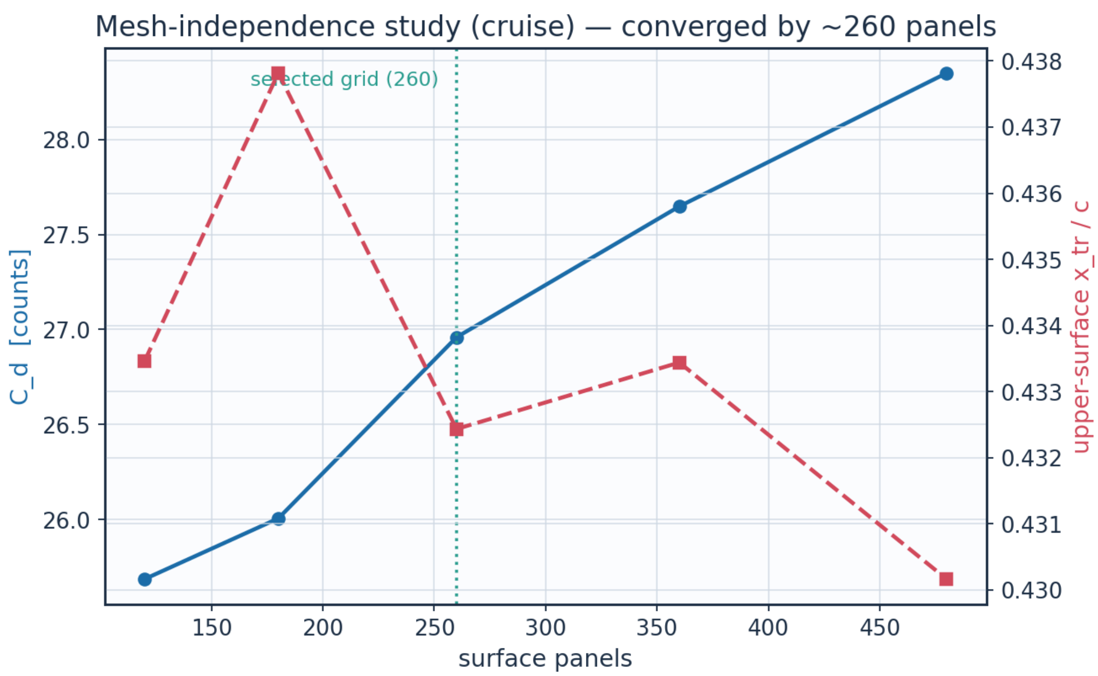


Fig. 11. Mesh-independence convergence.

layer	y_m	y_plus	cell_dy_m	growth_ratio
0.0	0.0	0.0	6.5127391337287455e-06	1.0
2.0	1.380700696350494e-	1.696	7.731923899562769e-	1.12

	05		06	
4.0	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6.0	5.2852109259255385e-05	6.4921512345600005	1.2166331946008715e-05	1.12
8.0	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10.0	0.0001142903336347	14.038988055628067	1.9143958857231197e-05	1.1199999999999997
12.0	0.0001571728014749	19.30650661697985	2.401418199051081e-05	1.12
14.0	0.0002109645691337	25.91408190033953	3.012338988889677e-05	1.12
16.0	0.0002784409624848	34.20262433578592	3.778678027663213e-05	1.120000000000001
18.0	0.0003630833503045	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20.0	0.0004692587615855	57.64195395516631	5.945823031734682e-05	1.12
22.0	0.0006024451974963	74.00206704136062	7.458440411007986e-05	1.12
24.0	0.0007695142627029	94.52419289668278	9.355867651568416e-05	1.12
26.0	0.000979085698098	120.26714756959888	0.0001173600038212	1.1200000000000006
28.0	0.0012419721066577	152.55910991130486	0.0001472163887934	1.1200000000000008
30.0	0.0015717368175549	193.0661474727408	0.0001846682381024	1.1199999999999997
32.0	0.0019853936709044	243.87817538980616	0.0002316478378757	1.1200000000000003
34.0	0.002504284827746	307.6167832089729	0.0002905790478312	1.119999999999999
36.0	0.0031551818948881	387.57049285733575	0.0003645023575995	1.12
38.0	0.0039716671759111	487.864426240242	0.0004572317573729	1.12
40.0	0.0049958663124264	613.6731362757596	0.0005735515164485	1.1200000000000014
42.0	0.0062806217092712	771.4875821443129	0.000719463022233	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism	laminar_run_pct	Cf_te
CRUISE	upper	0.457	0.432	2930000.0	1233.1	TS-natural	43.2	0.00016
CRUISE	lower	0.587	0.578	3766000.0	1298.2	TS-natural	57.8	0.00021
CLIMB	upper	0.039	0.018	421300.0	480.8	TS-natural	1.8	0.00012
CLIMB	lower	0.134	0.132	1431000.0	623.6	bypass	13.2	0.00016

Table 12. Transition prediction summary (transition_summary.csv).

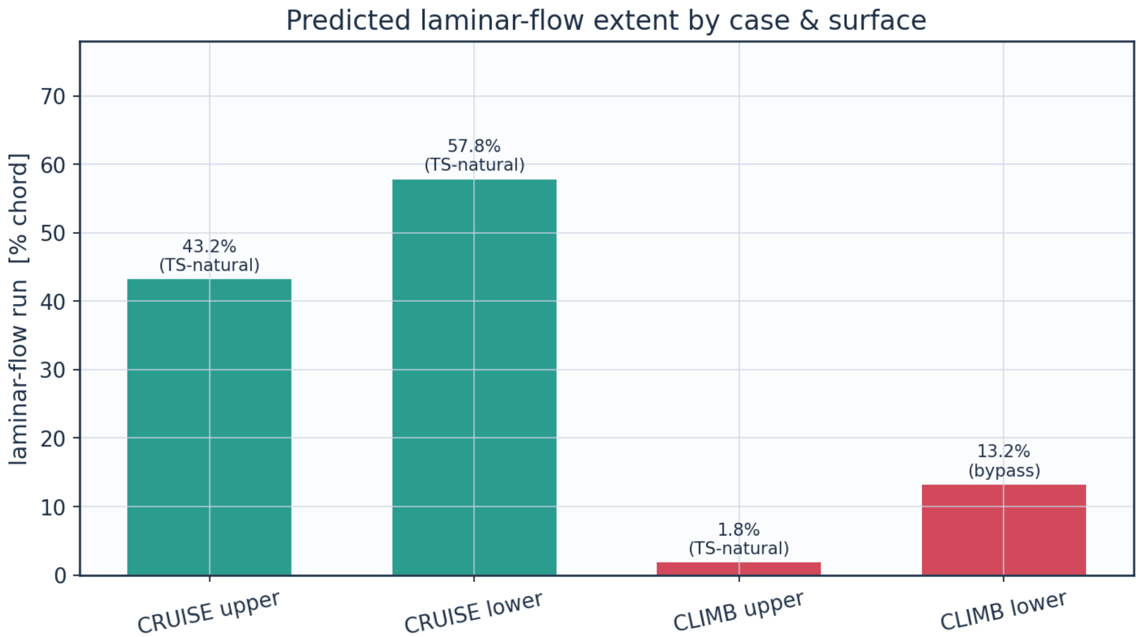


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

quantity	value	unit
Section lift coefficient Cl	0.5018	-
Section profile drag Cd	0.0027	-
Section Cd (counts)	27.0	counts
Section L/D	186.1	-
Wing lift (3D est, e=0.85)	41842.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Table 13. Integrated forces.

configuration	Cd_counts	mean_laminar_pct
NLF (UTSS predicted transition)	27.0	52.2
Fully turbulent (LE trip)	48.9	0.0
Viscous drag reduction	44.8	

Table 14. NLF vs fully-turbulent drag.

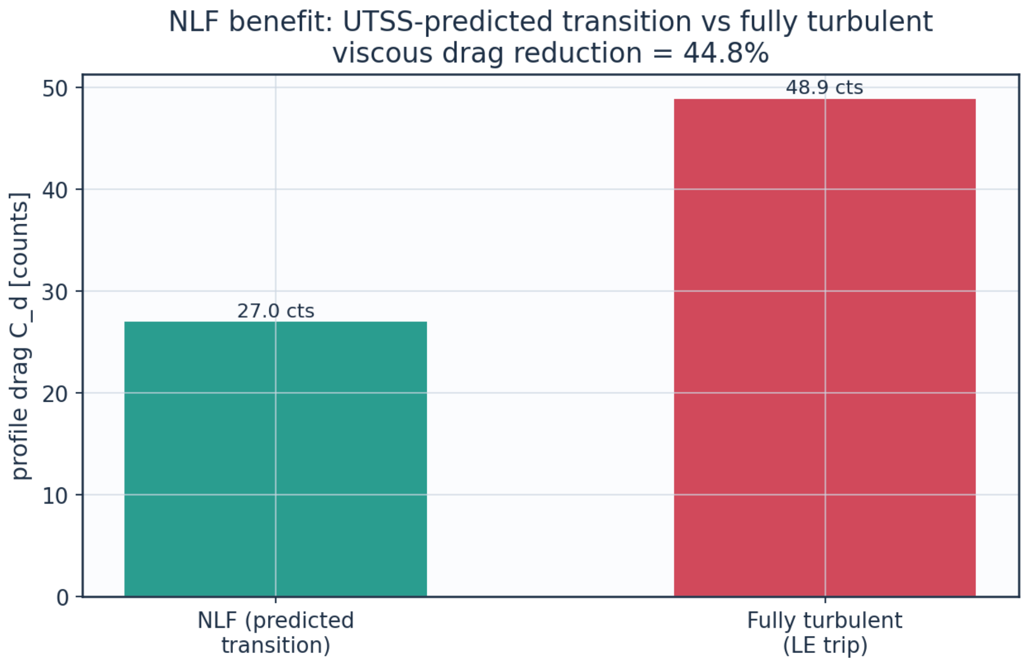


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

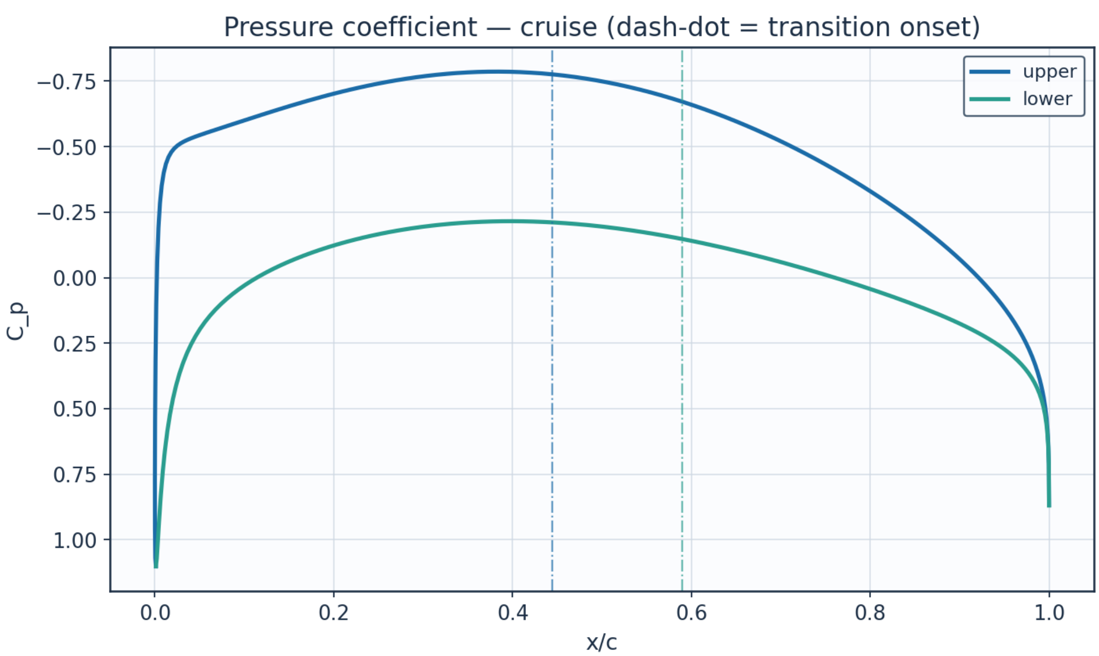


Fig. 13. Pressure coefficient C_p — cruise.

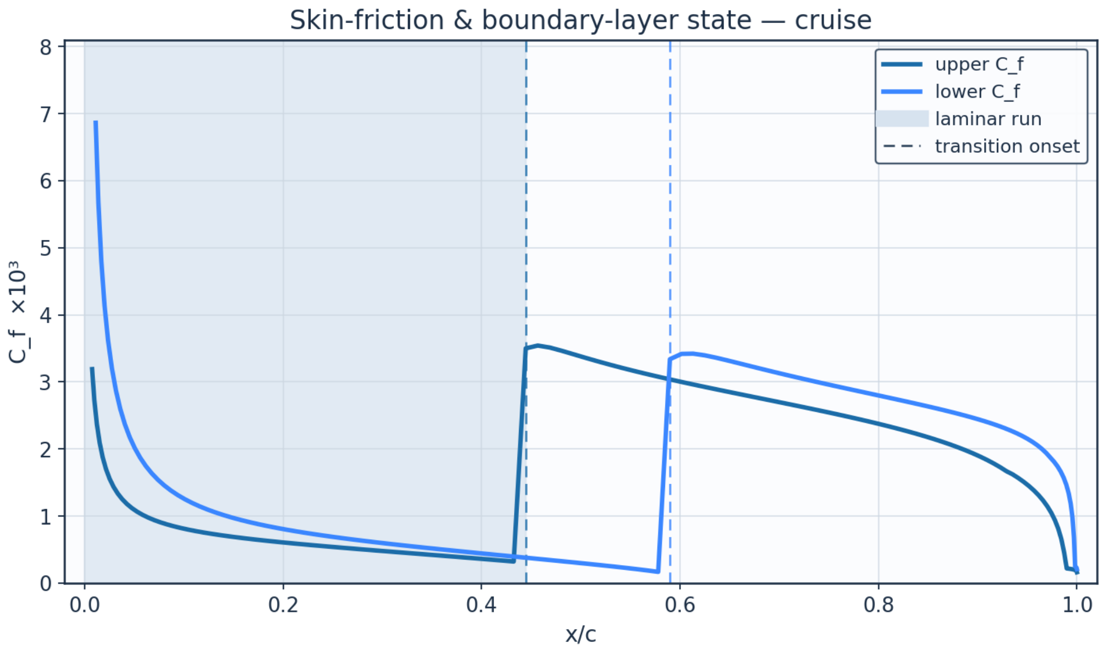


Fig. 14. Skin-friction C_f and laminar run — cruise.

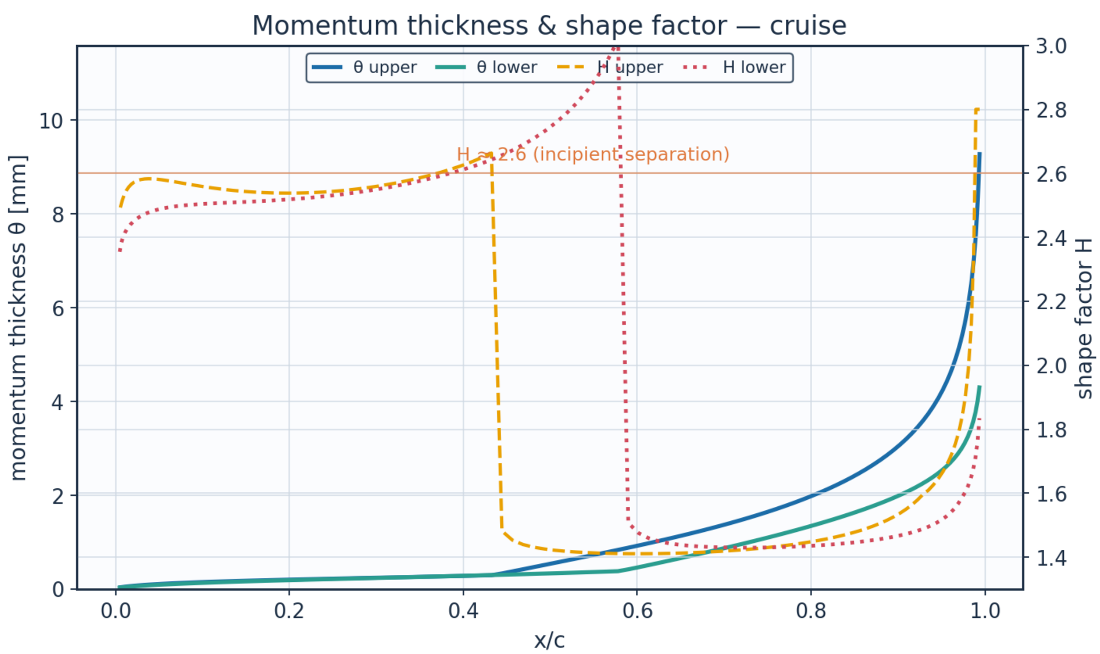


Fig. 15. Momentum thickness ϑ and shape factor H — cruise.

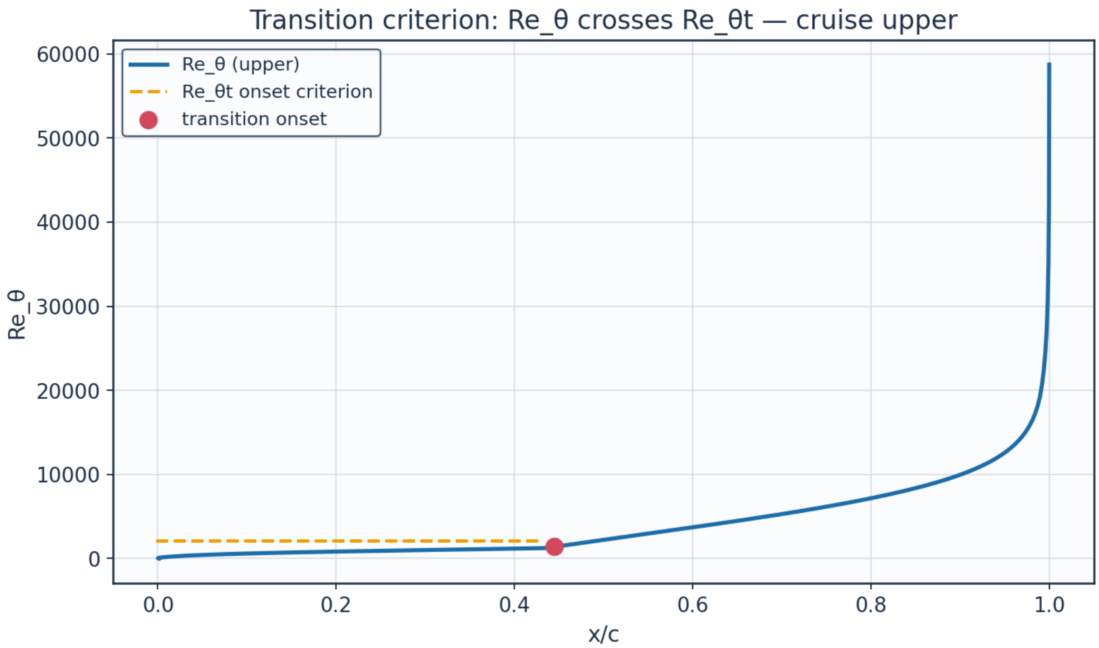


Fig. 16. Transition criterion Re_ϑ vs Re_ϑt — cruise.

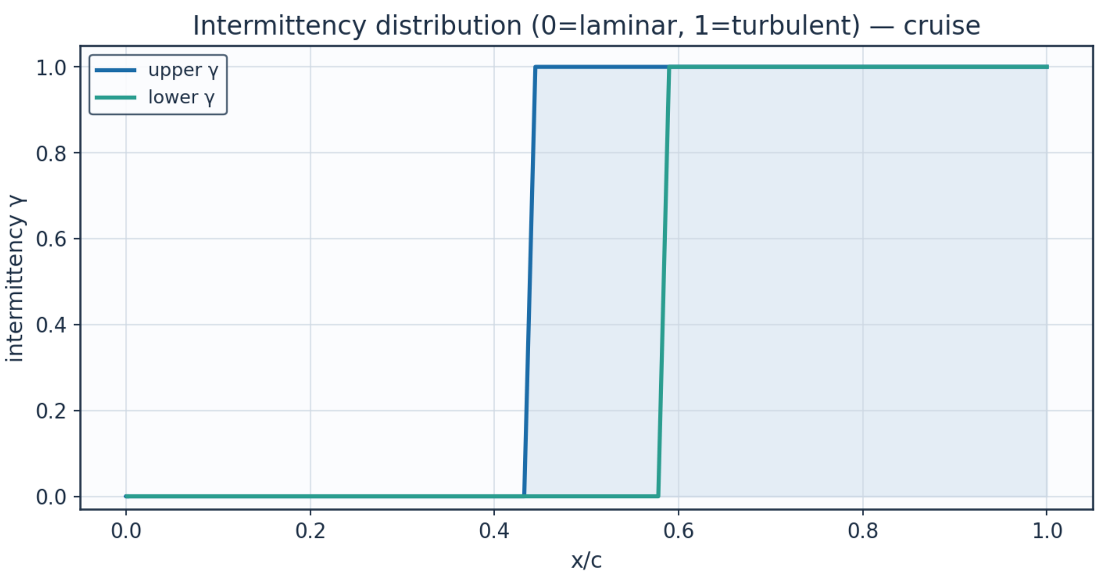


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011523200325028	0.0	0.0	1.1018664780939074	0.6743779759376758	0.0054416302825453	1e-05	2.6099999453597404	2549.4295521631888	0.0001725876462751	2071.8483164951444	0.0	laminar
0.000242727	0.014097598	44712.17719	0.545321836	88.07774049	0.710709004	0.023726573	2.357156794	0.012269311	53.48198503	2071.848316	0.0	lami

1046013	6799438	636864	2263359	458249	4378694	617726	6326266	8338141	165703	4951444		nar
0.0039984634929024	0.0303574508340593	96282.19328257376	-0.1889934015849139	134.13696378207896	1.0823659582155565	0.0352857650790395	2.469736760966773	0.0046122221795125	121.13073085887146	2071.8483164951444	0.0	laminar
0.0123828259304446	0.052871121292092	167686.92296127492	-0.4342862195904694	146.32725910183973	1.1807307959357485	0.0556075338763743	2.545485299945595	0.0023735313465547	208.24050116343795	2071.8483164951444	0.0	laminar
0.0253172774881581	0.0831882375254351	263841.19035694306	-0.5041257659832884	149.6165255093602	1.2072722494368586	0.078490183213124	2.5775438956420533	0.0015546621939321	300.5392756212738	2071.8483164951444	0.0	laminar
0.0426830314241258	0.1217109964051874	386020.6097196828	-0.5346198063882681	151.03025162861334	1.218679761449772	0.1013877008407719	2.5827502987046134	0.0011811190399166	391.8822095845795	2071.8483164951444	0.0	laminar
0.0643227586523211	0.1683869080802935	534058.7033695233	-0.5602079825700778	152.20640645138076	1.2281702844634186	0.1232764382036634	2.5752582410709937	0.0009770155940745	480.196884305317	2071.8483164951444	0.0	laminar
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.5889179272140977	153.5153258571104	1.2387320995430495	0.1438100654922687	2.563274868453051	0.000848234433964	564.9985721480421	2071.8483164951445	0.0	laminar
0.1196122456699886	0.2849313998941607	903693.1416555132	-0.6215827965372597	154.99111243577204	1.250640384249194	0.1629878673543141	2.551450446445226	0.0007567308656733	646.4998076476546	2071.8483164951444	0.0	laminar
0.1527694018013956	0.353848290318842	1122271.0914502463	-0.6567690814927153	156.56525350268606	1.263342302171124	0.1810095226692608	2.542533164062804	0.0006849269121255	725.2757153438041	2071.8483164951445	0.0	laminar
0.1892179872734863	0.4290798210019653	1360876.6587546526	-0.6922076946556974	158.13484523358863	1.2760075109989126	0.1981991276343066	2.5382750307941677	0.000623805195057	802.1131640663597	2071.8483164951444	0.0	laminar
0.2286307839316745	0.5099588956971477	1617394.0700776232	-0.7253026000789115	159.5866938991054	1.2877226380431552	0.2149585127496167	2.5399584156338757	0.000568315064827	877.9254801051701	2071.8483164951444	0.0	laminar
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7533932601692048	160.80872570370516	1.2975833474841738	0.2317360799163555	2.5487477216506247	0.0005153731419897	953.6951938170474	2071.8483164951444	0.0	laminar
0.3148902862023041	0.6857636986922375	2174979.491676561	-0.77395958212237	161.69756919660242	1.3047555236820514	0.2490060834747701	2.565969510539149	0.0004628932471579	1030.433041355294	2071.8483164951444	0.0	laminar

			18									
0.360937646562712	0.7791556260346346	2471182.8734613145	-0.784812690647283	162.16465934647948	1.3085245256289428	0.2672560850594751	2.593386420394073	0.0004092565510075	1109.1496431580938	2071.848316495145	0.0	laminar
0.4083553901730684	0.875139546452957	2775607.063366206	-0.7842649860650261	162.1411197751582	1.3083345822311592	0.2869812191720717	2.634417567799684	0.000352977041816	1190.8387870313256	2071.8483164951444	0.0	laminar
0.4566864473674744	0.9728858705600144	3085620.9219665444	-0.7712569098159315	161.5810430777686	1.3038152615744822	0.3794978641116916	1.4530682004599045	0.0035424655887303	1569.3003468692243		1.0	turbulent
0.5054586679985256	1.071544176368635	3398527.2368188924	-0.7454187088381593	160.4627537071277	1.2947916612775314	0.5614437418266487	1.4215374686511746	0.0033566482830954	2305.6151309564643		1.0	turbulent
0.5541908676652394	1.1702450637414947	3711568.604064828	-0.707056278364234	158.78788331631787	1.2812769473290269	0.7451701811423619	1.4129362173259858	0.0031623383753583	3028.1623447317884		1.0	turbulent
0.6023997176209082	1.268103309912791	4021937.436535685	-0.6570650024969856	156.57842511702526	1.263448585380267	0.934681811549487	1.410792701692836	0.0029972233597335	3745.43306029041		1.0	turbulent
0.649607126440503	1.3642231699528122	4326792.774793668	-0.5967887204991281	153.87221973840155	1.2416119156423058	1.1345975550585343	1.4127557264169894	0.0028500841054955	4467.9508779230055		1.0	turbulent
0.6953476717367839	1.4577062094846656	4623285.129505811	-0.5278484305271463	150.71747104689732	1.2161559004311917	1.3494404742176995	1.418230941444463	0.0027125202468977	5205.03504211194		1.0	turbulent
0.7391756157699833	1.547661543814744	4908588.956039555	-0.4519657541873016	147.16687039720318	1.187505712100625	1.5840063924966776	1.4272273857066091	0.0025786624253737	5965.863491072978		1.0	turbulent
0.7806710851424759	1.633217905018701	5179940.926632257	-0.3708007619632745	143.2717466982367	1.1560755292790632	1.8438344312972792	1.4401643844649847	0.0024438055965749	6760.6554586470975		1.0	turbulent
0.8194451029680374	1.7135366561310357	5434681.206410178	-0.2858150479089136	139.0764000663424	1.122222814492303	2.1358646116714968	1.4579244509587703	0.0023034148082671	7602.098425830968		1.0	turbulent
0.855143310574868	1.787824762773985	5670294.59442127	-0.1981612368526684	134.61243710955	1.0862026049472524	2.4694843170092	1.4820947951539356	0.0021522317263054	8507.41829669423		1.0	turbulent

0.8874483761131826	1.855346811455994	5884448.641069436	-0.1085902183180816	129.89245152018606	1.0481165205365803	2.85835655161658	1.5155402855423157	0.0019831091267138	9501.816580522012		1.0	turbulent
0.916081229913384	1.9154353854551616	6075026.556438111	-0.0173555837833283	124.90156875556713	1.0078445369344735	3.323907709805697	1.5638014092374797	0.0017849380191167	10624.859050187744		1.0	turbulent
0.9408013677638524	1.9674994072259613	6240153.669205437	0.0759256832053258	119.58356245557384	0.9649329574384596	3.9025525747010983	1.6263925108982584	0.0015688125758713	11943.361484254716		1.0	turbulent
0.9614065112832176	2.0110303614800173	6378217.163870198	0.1725468608702507	113.81345518080396	0.9183733253029868	4.660583068937207	1.719678349206924	0.0013104450533975	13575.010948508896		1.0	turbulent

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	**arc_s_m**	**Re_x**	**Cp**	**Ue_ms**	**Ue_Uinf**	**theta_mm**	**H_shape**	**Cf**	**Re_theta**	**Re_theta_trans**	**intermittency_gamma**	**state**
0.0011523200325028	0.0	0.0	1.1018664780939074	0.6743779759376758	0.0054416302825453	1e-05	2.609999951526142	2549.4295222460187	0.0001725876462751	2071.8483164951444	0.0	laminar
0.0067207749723691	0.0175589809676828	55690.354523400645	0.8272660671741117	61.86995831428995	0.4992355188864786	0.0360998768098014	2.383367050372955	0.0110788262099739	57.15995401392783	2071.8483164951444	0.0	laminar
0.0168974815052617	0.0419271025953991	132976.692210909	0.5213997099578502	89.95065248466172	0.7258217378980429	0.0529461853074648	2.439017669829458	0.0048001190392807	121.88361332908732	2071.8483164951444	0.0	laminar
0.031587190340849	0.0742427608527724	235469.5685527914	0.3249255383027969	104.06544700919068	0.8397155720046309	0.0729648924458002	2.469823338295471	0.0028746178215667	194.3241730005226	2071.8483164951444	0.0	laminar
0.0506501048346889	0.1147428056677419	363920.1806988166	0.1950002508665796	112.43016773262164	0.9072114262848172	0.0937924906661417	2.488039615326091	0.0020122165347153	269.8716440330299	2071.848316495145	0.0	laminar
0.0739026679245839	0.1632868568057734	517883.2963747615	0.1010103979092831	118.11262634139554	0.9530638116653156	0.114584947642155	2.4984264887283723	0.0015422773765276	346.3619887458968	2071.848316495144	0.0	laminar
0.1011191421095574	0.2195305250490136	696266.645039893	0.0275361185138918	122.37112447797548	0.9874261029100077	0.1349744515350473	2.504408510497892	0.0012516884846285	422.7044789914756	2071.8483164951444	0.0	laminar
0.1320340524549544	0.2829946292313039	897550.4477803087	-0.0328993579368748	125.76587055281448	1.0148186834828097	0.1548505072578215	2.50858154641777	0.0010544558188138	498.40427770690434	2071.8483164951444	0.0	laminar
0.1663	0.3531	11199	-	128.55	1.0373	0.1742	2.5129	0.0009	573.34	2071.8	0.0	la

455116316067	01330536502	02.021447278	0.0837640199419475	355423838213	128106678653	722652355156	95255388638	100822058467	85152818953	483164951444		minar
0.2037193715819191	0.4291984666370509	1361252.9572136996	-0.1264284129239836	130.84602165898758	1.0558109831806464	0.193419376811025	2.519345577877747	0.0007972991628095	647.6894598935661	2071.8483164951444	0.0	laminar
0.2437940563058938	0.5105788670995084	1619360.3816338223	-0.1611531100556272	132.68263393649002	1.0706308102555977	0.2125576451808114	2.5291241637600383	0.0007039808429943	721.7671995485019	2071.8483164951444	0.0	laminar
0.2861858395053481	0.5964965491145282	1891858.323288388	-0.187658005003926	134.06756513308085	1.0818059728610068	0.2320118602543016	2.5437485485588143	0.0006227416068362	796.0497965256656	2071.8483164951444	0.0	laminar
0.3304942587938113	0.6861799377514877	2176299.6423629136	-0.2054902294096907	134.99133760688824	1.089259994859075	0.2521428378381306	2.564683607082118	0.00054881652269	871.0816490204636	2071.848316495145	0.0	laminar
0.3763073240029427	0.7788417676108137	2470187.43489202	-0.2142792010368094	135.44431963547697	1.0929151568189388	0.2733271127026738	2.593561060617944	0.0004789560916819	947.4359005165832	2071.8483164951444	0.0	laminar
0.4232061945767489	0.8736854550834547	2770995.2431229902	-0.213909854122899	135.4253140603062	1.092761798736266	0.2959383228694588	2.6331137230267982	0.0004108116889601	1025.6693721653007	2071.8483164951444	0.0	laminar
0.4707690743351984	0.9699081665724224	3076176.7867379705	-0.2046212723147929	134.94646911402558	1.088897946040452	0.3203298398255593	2.6923104986956523	0.0003424388578886	1106.2804805874682	2071.848316495145	0.0	laminar
0.5185741934072935	1.0667013259415112	3383167.5722867246	-0.1870323400609301	134.03503786013175	1.0815435067073007	0.3468188897752094	2.787138865126507	0.0002713245237586	1189.6724718165433	2071.8483164951444	0.0	laminar
0.5662018980247188	1.1632497261198471	3689382.075536924	-0.1621004751380548	132.73238458462728	1.0710322537237111	0.3756733234164355	2.954516085058231	0.0001919786027508	1276.1260701804902	2071.8483164951444	0.0	laminar
0.6132360223540745	1.2587306107483358	3992210.828808826	-0.1310277122034001	131.0907600194213	1.0577858040095516	0.5130345414157332	1.4594357491111023	0.0034216577339982	1721.1749428207409		1.0	turbulent
0.659264843410772	1.352314020522654	4289021.519442441	-0.0951351098104894	129.16852904093932	1.042275110191281	0.7001282151379985	1.4359356706461828	0.003278692112256	2314.411728018191		1.0	turbulent
0.703881995	1.443165382	4577167.201	-0.0557	127.0245984	1.024975497	0.889874195	1.430191550	0.003116238	2892.829060		1.0	turbu

0989186	4649215	974402	282947300742	3724029	6401104	9042816	3756517	8181889	377729			lent
0.7466877251231454	1.5304508477617995	4854003.2277135777	-0.0139732207381602	124.71270106297332	1.006320542687799	1.0831930966347862	1.4302334579084968	0.0029707019850141	3457.187199714529		1.0	turbulent
0.7872908170545605	1.613345372850863	5116912.874848304	0.0292065479862741	122.27595559534484	0.9866581746974072	1.2815041477979248	1.4337311261245251	0.0028393343592244	4010.2127104953584		1.0	turbulent
0.8253113815520968	1.6910430987466305	5363340.267688533	0.0732090480954794	119.74177812267357	0.9662096171100124	1.486273943940221	1.4400824875953482	0.0027170294186148	4554.606845952289		1.0	turbulent
0.8603845694278387	1.7627692889369593	5590828.239095532	0.1178167070088907	117.11679090307612	0.9450283056568992	1.6996025590872912	1.4493952007532325	0.002598661022575	5094.163357346465		1.0	turbulent
0.892165104533994	1.827792963070222	5797058.399693135	0.163258645360086	114.38078351124722	0.9229511601867452	1.9252467263353392	1.4623932732940246	0.0024784433496168	5635.673028240077		1.0	turbulent
0.9203324048391728	1.8854393957576685	5979890.779276088	0.2103002435852176	111.47774743784164	0.8995262418590072	2.1703653496110213	1.4807044409940366	0.0023485930092237	6191.94883762199		1.0	turbulent
0.9445959764010288	1.9351017924661469	6137400.858264567	0.2604122077833569	108.29966444597052	0.8738819396040853	2.448868158435427	1.5077207074194332	0.0021968741446794	6787.328087938799		1.0	turbulent
0.9647007361335488	1.9762516523275704	6267912.434562116	0.3161381027017609	104.6522734793462	0.8444507395286849	2.7889416420564648	1.5511172946948106	0.0020009428112135	7469.549707902191		1.0	turbulent
0.980431941297911	2.008447529886631	6370025.412459643	0.3819991826500151	100.17042017489692	0.8082861707947988	3.2536919091000613	1.6213787356202185	0.0017408155655174	8341.079302621814		1.0	turbulent

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated Tu)

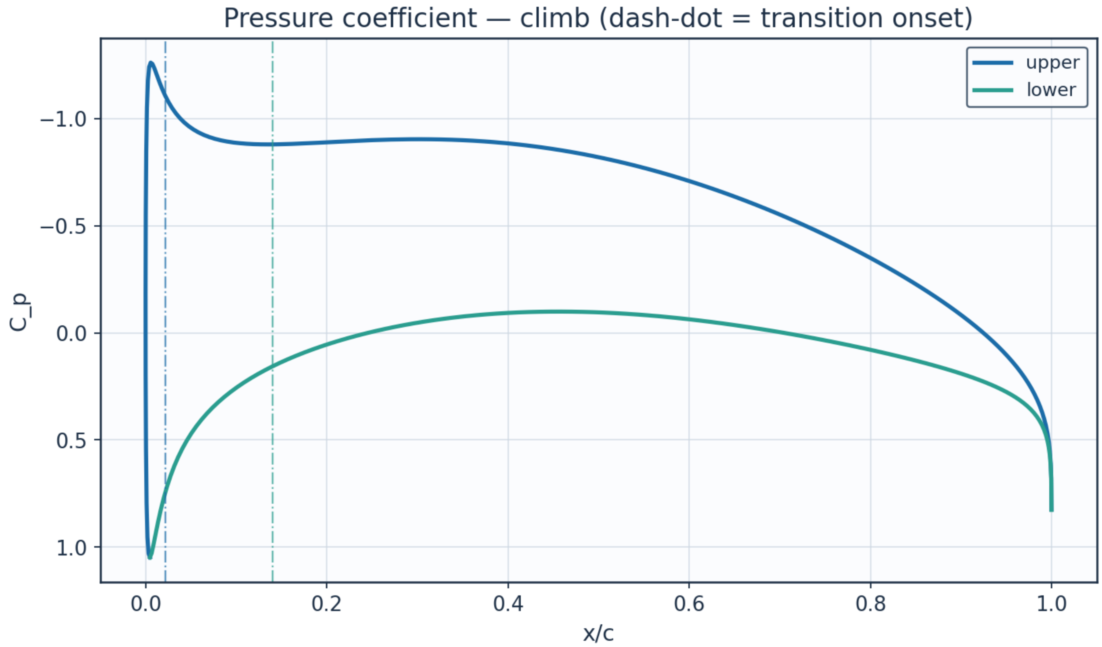


Fig. 18. Pressure coefficient C_p — climb.

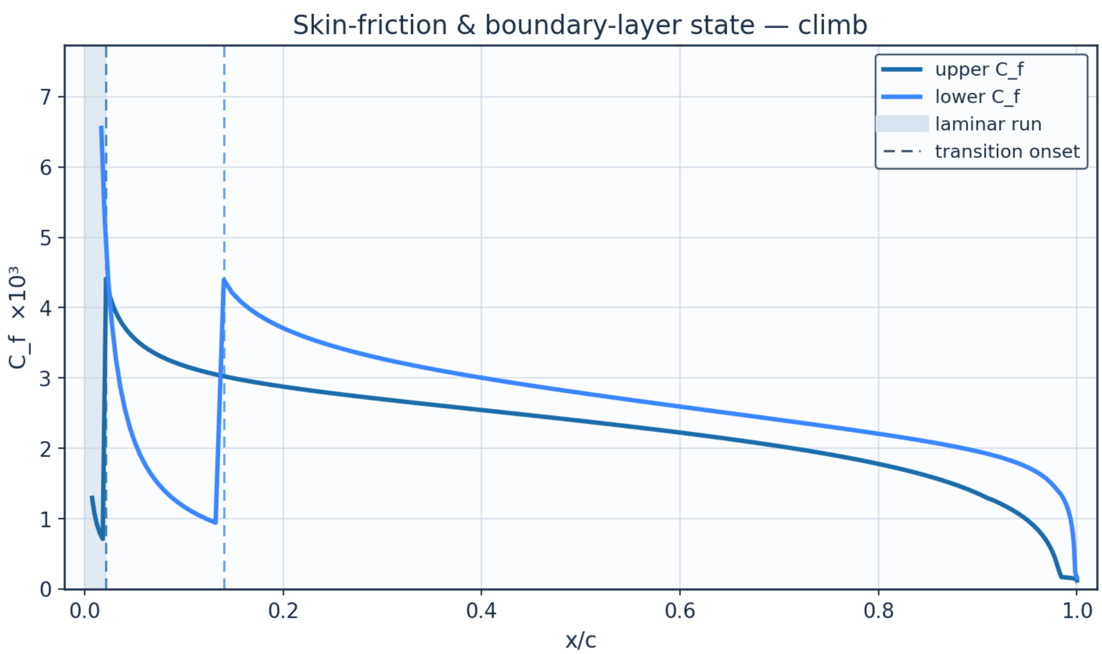


Fig. 19. Skin-friction C_f and laminar run — climb.

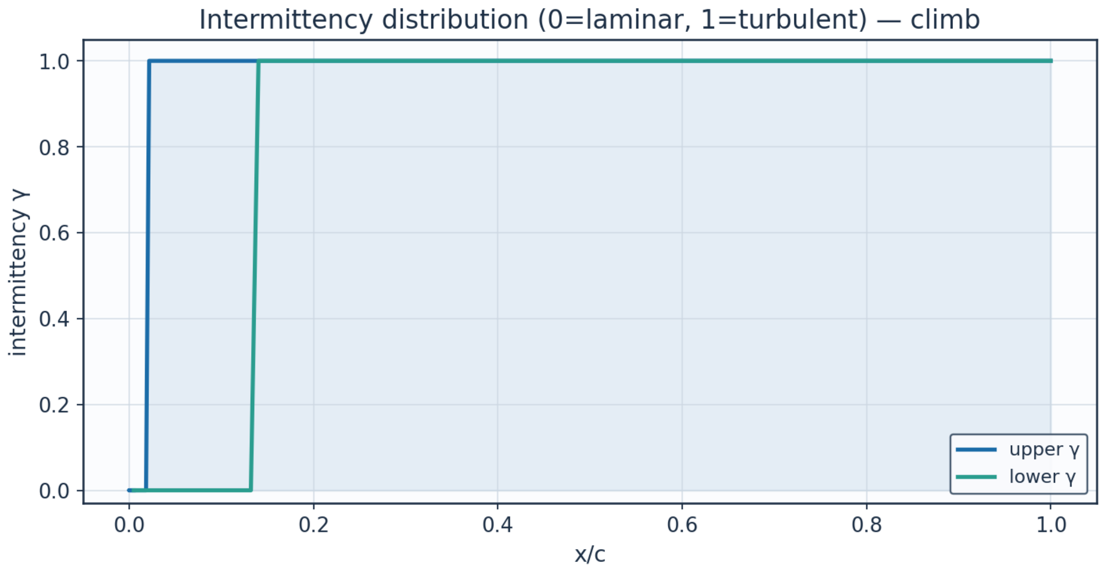


Fig. 20. Intermittency γ — climb.

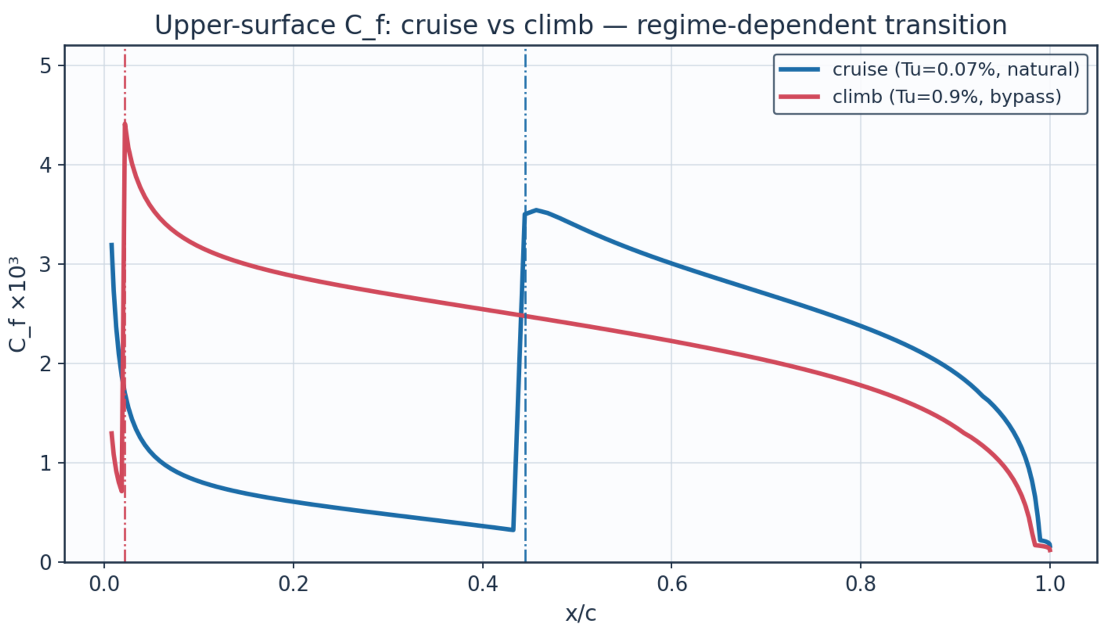


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1086	0.00281	-38.6	0.661	0.185
-2.0	0.0271	0.00251	10.8	0.638	0.308
-1.0	0.1628	0.0023	70.7	0.614	0.423
0.0	0.2985	0.00231	129.1	0.566	0.495
1.0	0.4341	0.00253	171.9	0.481	0.554
2.0	0.5695	0.00294	193.5	0.373	0.59
3.0	0.7048	0.00363	194.3	0.219	0.625

4.0	0.8399	0.0044	190.8	0.077	0.659
5.0	0.9746	0.00497	196.2	0.029	0.682
6.0	1.1091	0.00546	203.0	0.015	0.715
7.0	1.2432	0.00602	206.4	0.01	0.736
8.0	1.3769	0.00665	207.2	0.008	0.757

Table 17. Aerodynamic polar (aero_polar.csv).

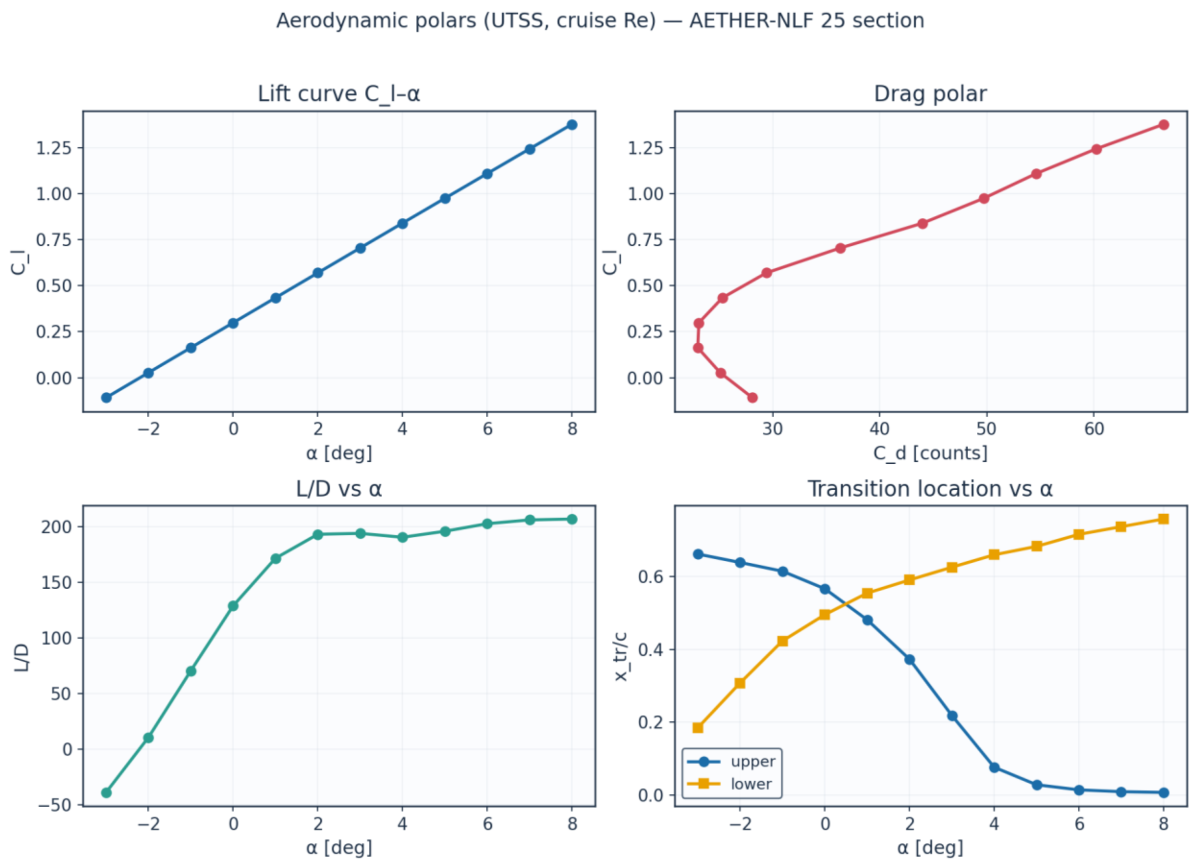


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.0	2.6	8246200.0	1.5	0.373	0.542	0.00279	0.458
0.089	0.757	2.484	7878900.0	1.23	0.42	0.542	0.00263	0.481
0.178	1.515	2.368	7511500.0	0.97	0.457	0.531	0.00254	0.494
0.267	2.272	2.253	7144200.0	0.7	0.493	0.519	0.00246	0.506
0.356	3.029	2.137	6776900.0	0.43	0.518	0.519	0.0024	0.518
0.445	3.786	2.021	6409500.0	0.16	0.554	0.507	0.00233	0.53
0.535	4.544	1.905	6042200.0	-0.1	0.578	0.495	0.00231	0.537
0.624	5.301	1.789	5674900.0	-0.37	0.602	0.495	0.00228	0.549
0.713	6.058	1.673	5307600.0	-0.64	0.614	0.483	0.0023	0.549
0.802	6.815	1.558	4940200.0	-0.91	0.626	0.471	0.00234	0.548
0.891	7.573	1.442	4572900.0	-1.17	0.638	0.459	0.00238	0.548
0.98	8.33	1.326	4205600.0	-1.44	0.65	0.459	0.0024	0.554

Table 18. Span-wise distribution (spanwise_distribution.csv).

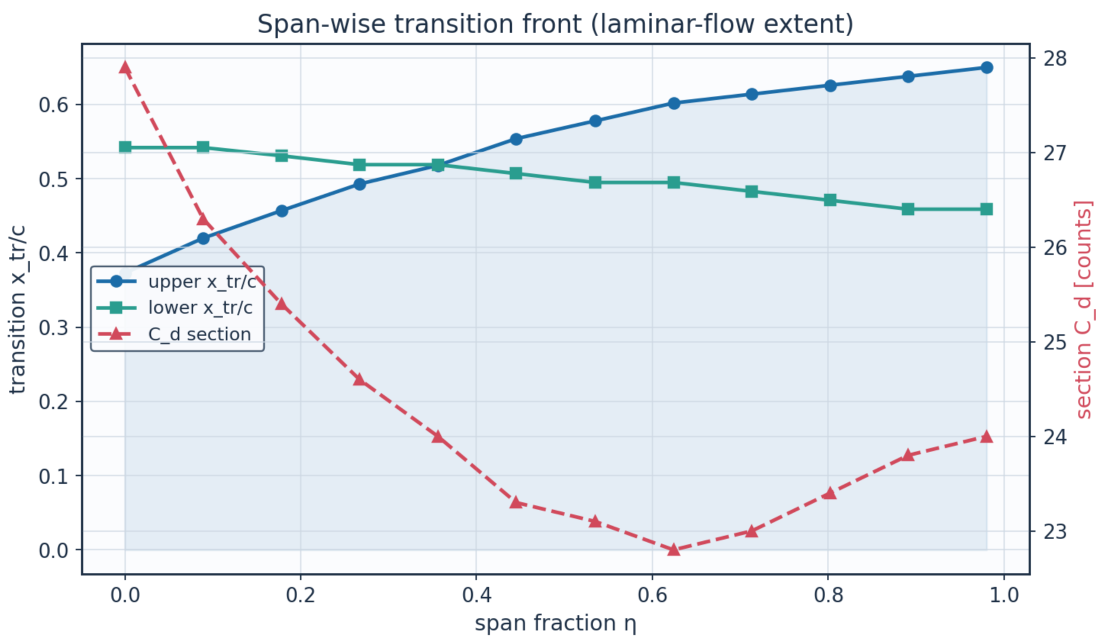


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

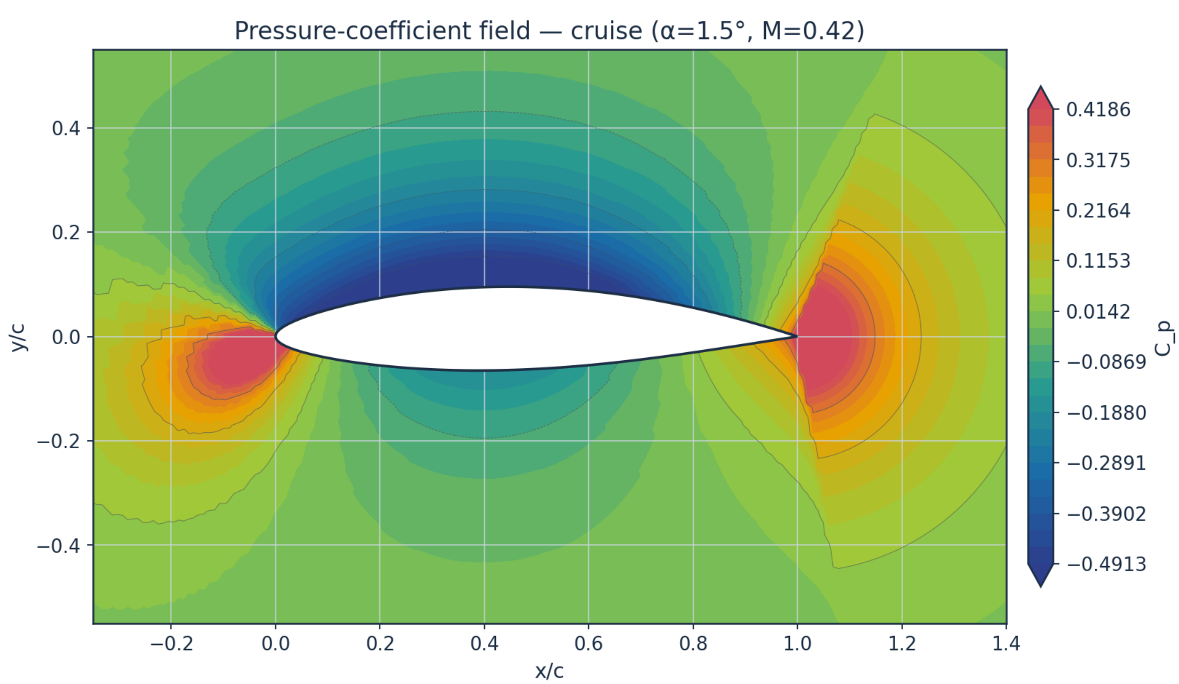


Fig. 24. Pressure-coefficient contour — cruise.

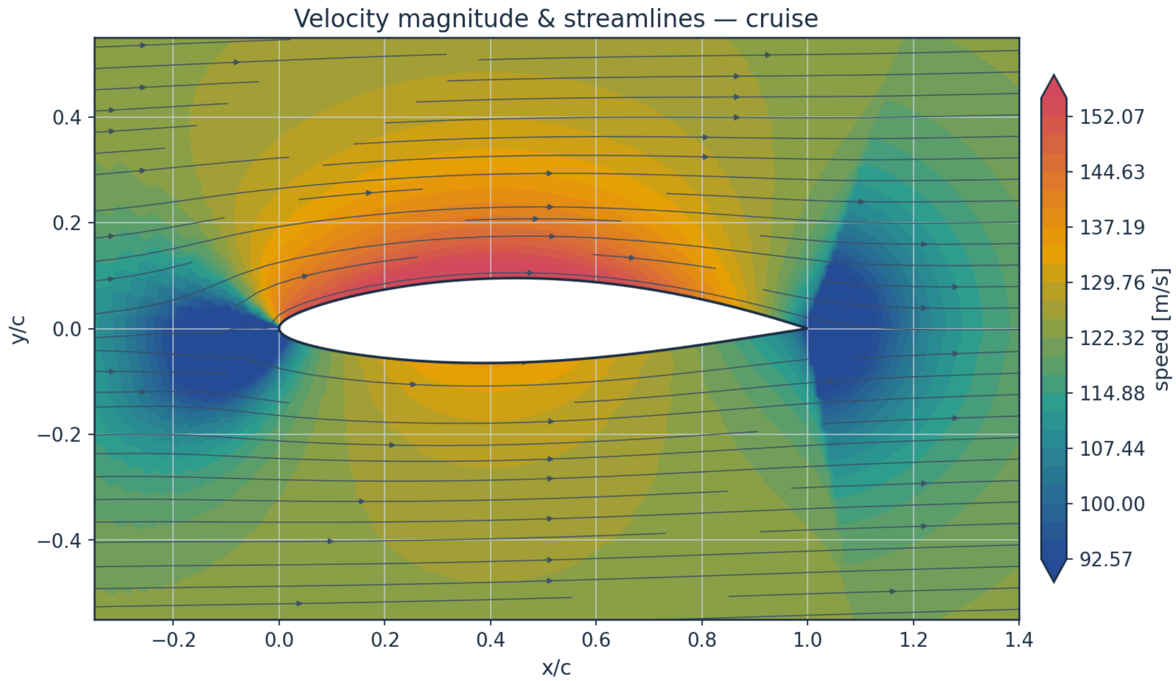


Fig. 25. Velocity magnitude and streamlines — cruise.

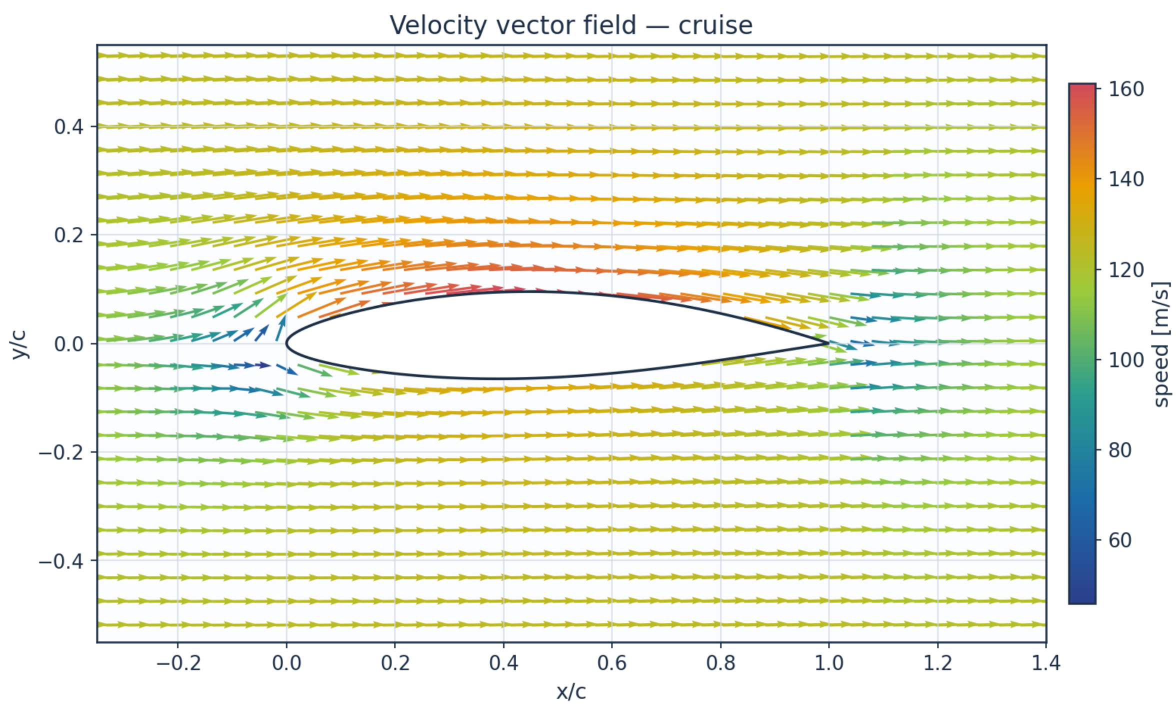


Fig. 26. Velocity vector field — cruise.

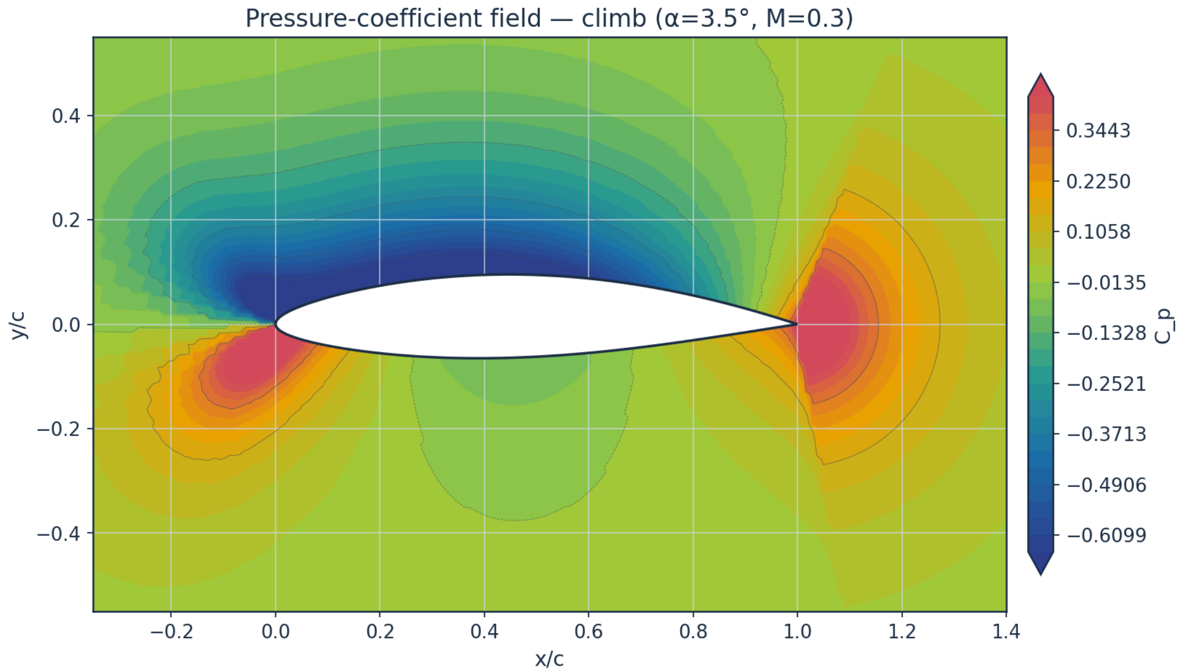


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

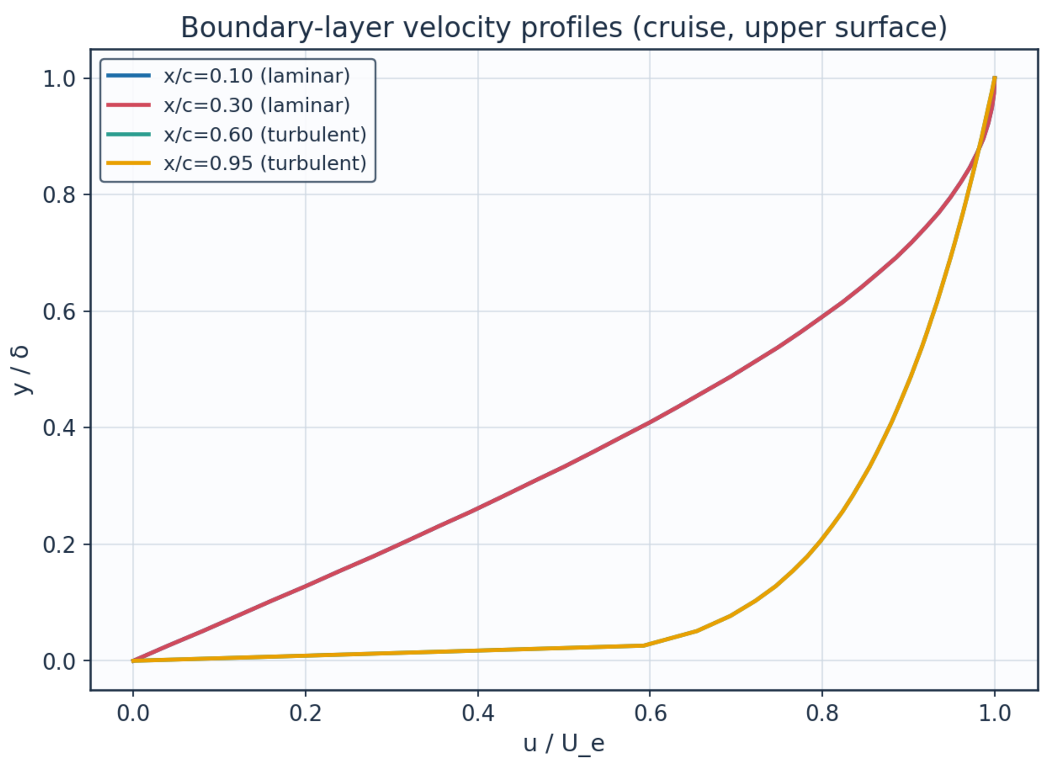


Fig. 28. Boundary-layer velocity profiles.

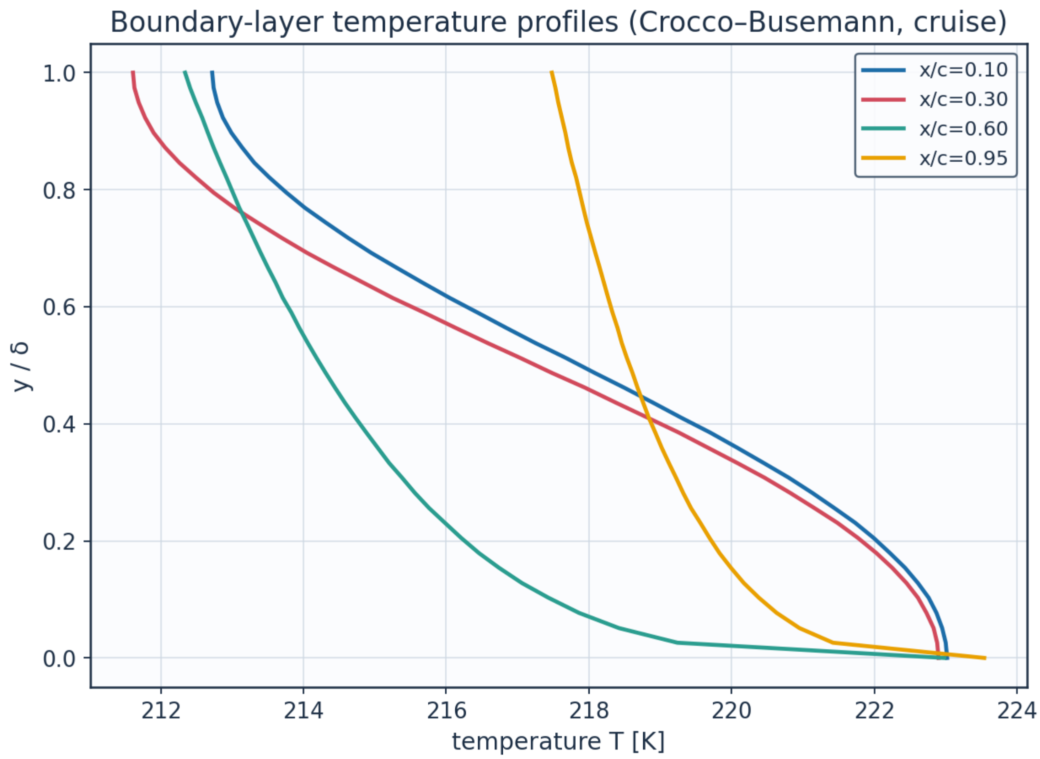


Fig. 29. Boundary-layer temperature profiles (K).

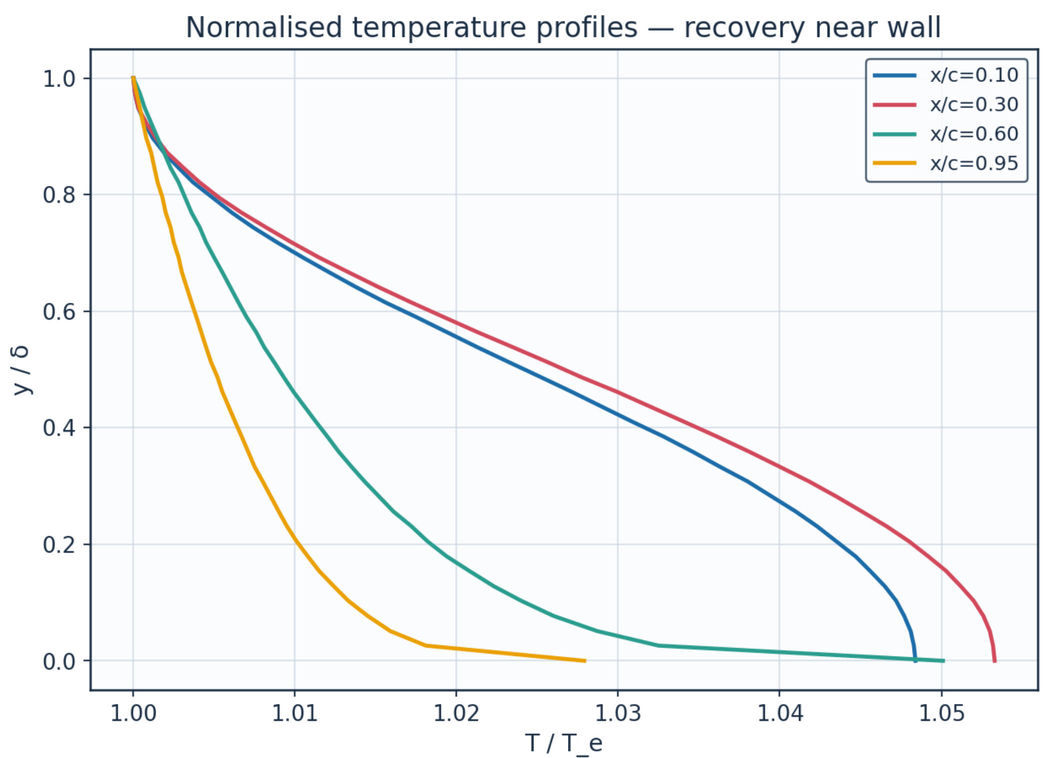


Fig. 30. Normalised temperature profiles T/T_e.

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K	Me_edge	state
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02	0.521	laminar
x/c=0.10	0.1	0.128	0.2441	0.2	1.0465	222.61	0.521	laminar
x/c=0.10	0.1	0.256	0.4881	0.392	1.041	221.44	0.521	laminar
x/c=0.10	0.1	0.385	0.7322	0.5681	1.0328	219.7	0.521	laminar
x/c=0.10	0.1	0.513	0.9763	0.7212	1.0232	217.67	0.521	laminar
x/c=0.10	0.1	0.641	1.2203	0.8452	1.0138	215.67	0.521	laminar
x/c=0.10	0.1	0.769	1.4644	0.935	1.0061	214.02	0.521	laminar
x/c=0.10	0.1	0.897	1.7084	0.9871	1.0012	212.99	0.521	laminar
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9	0.547	laminar
x/c=0.30	0.3	0.128	0.4014	0.2	1.0512	222.45	0.547	laminar
x/c=0.30	0.3	0.256	0.8028	0.392	1.0451	221.16	0.547	laminar
x/c=0.30	0.3	0.385	1.2043	0.5681	1.0361	219.26	0.547	laminar
x/c=0.30	0.3	0.513	1.6057	0.7212	1.0256	217.03	0.547	laminar
x/c=0.30	0.3	0.641	2.0071	0.8452	1.0152	214.84	0.547	laminar
x/c=0.30	0.3	0.769	2.4085	0.935	1.0067	213.03	0.547	laminar
x/c=0.30	0.3	0.897	2.8099	0.9871	1.0014	211.9	0.547	laminar
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98	0.531	turbulent
x/c=0.60	0.6	0.128	1.3421	0.7457	1.0223	217.06	0.531	turbulent
x/c=0.60	0.6	0.256	2.6842	0.8233	1.0161	215.76	0.531	turbulent
x/c=0.60	0.6	0.385	4.0263	0.8724	1.012	214.88	0.531	turbulent
x/c=0.60	0.6	0.513	5.3684	0.909	1.0087	214.18	0.531	turbulent
x/c=0.60	0.6	0.641	6.7105	0.9384	1.006	213.61	0.531	turbulent
x/c=0.60	0.6	0.769	8.0526	0.9632	1.0036	213.1	0.531	turbulent
x/c=0.60	0.6	0.897	9.3948	0.9847	1.0015	212.66	0.531	turbulent
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54	0.396	turbulent
x/c=0.95	0.95	0.128	6.1068	0.7457	1.0124	220.17	0.396	turbulent
x/c=0.95	0.95	0.256	12.2136	0.8233	1.009	219.43	0.396	turbulent
x/c=0.95	0.95	0.385	18.3204	0.8724	1.0067	218.93	0.396	turbulent

Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

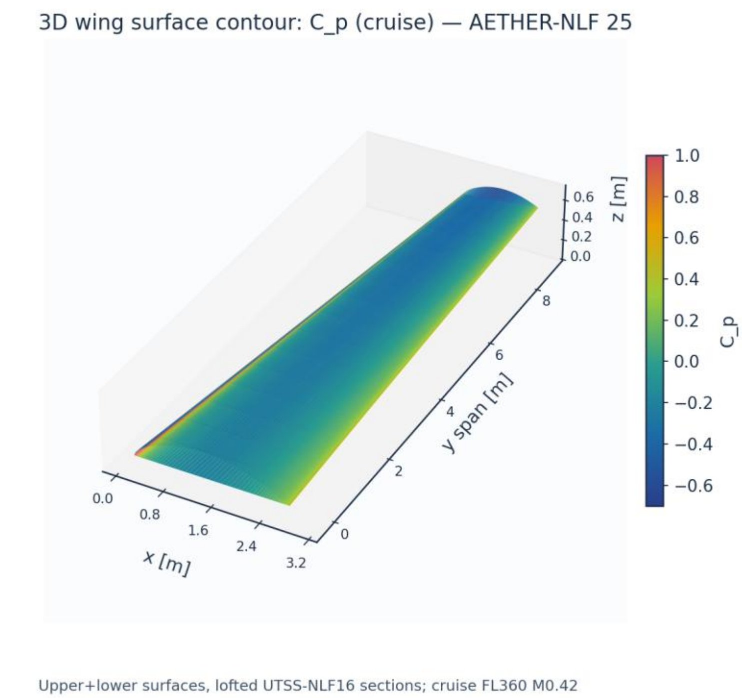


Fig. 31. 3-D wing surface contour — pressure C_p.

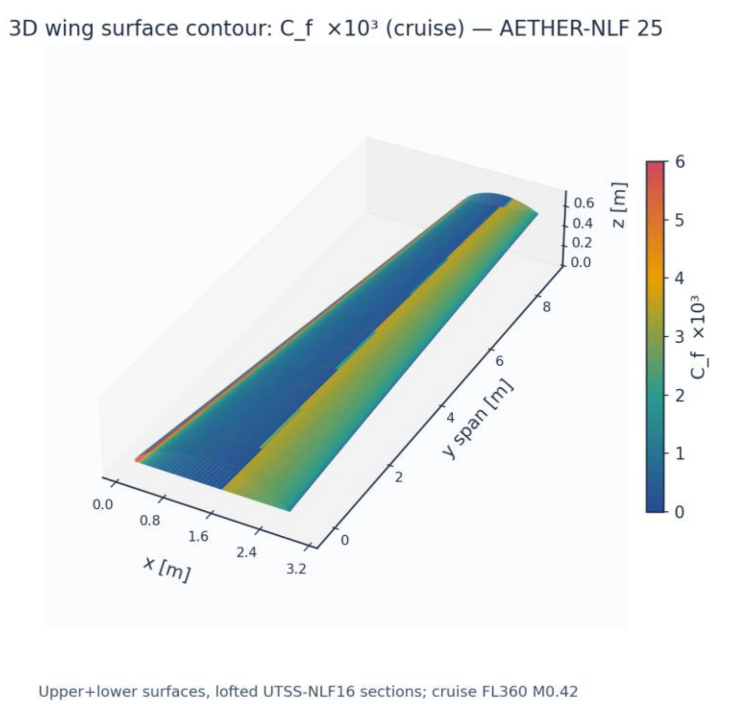


Fig. 32. 3-D wing surface contour — skin friction C_f (×10³).

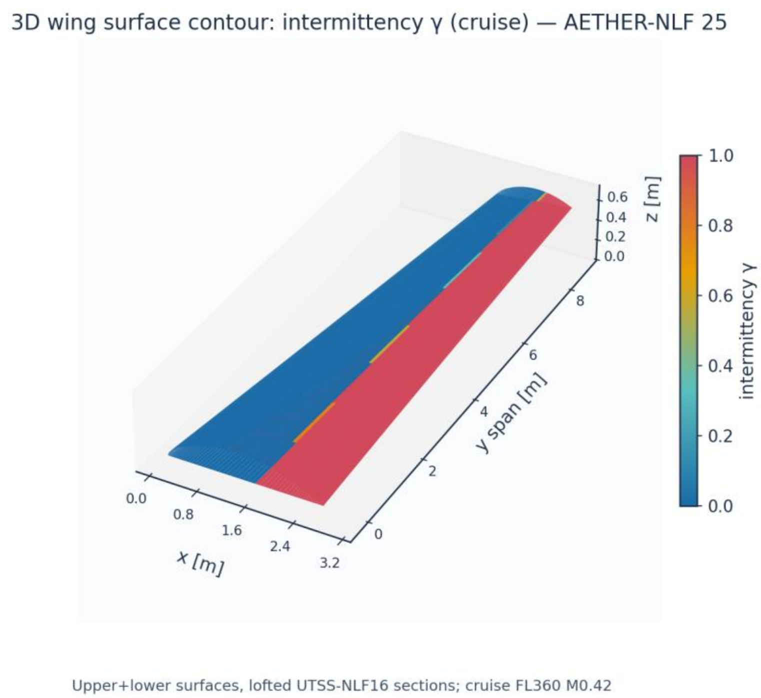


Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

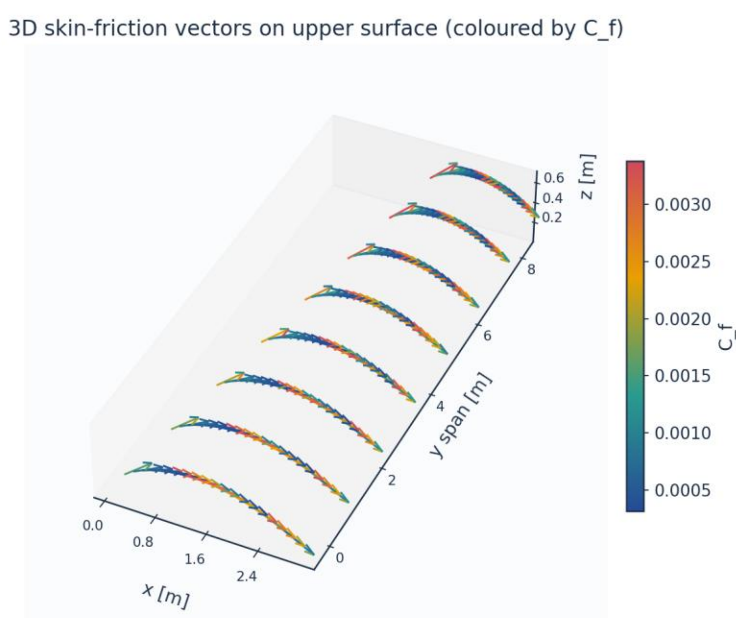


Fig. 34. 3-D skin-friction vectors on the upper surface.

11. Validation and Calibration

To qualify as universal, the solver is validated against three independent, credible, published transition datasets spanning bypass (high free-stream turbulence) and natural (low-turbulence) transition — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number Re_θt is the primary validation metric.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp	Re_x_tr_pred	mean_Cf_err_pct	Re_theta_meas_over_blasius	mechanism
ERCOFTAC T3A flat plate (bypass transition)	3.043	1.52999999999999998	272.3	394.5	44.9	134800.0	277100.0	20.4	1.117	bypass
ERCOFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	168.4	-7.1	59100.0	48650.0	14.5	1.123	bypass
Schubauer & Skramstad flat plate (natural transition)	0.03		1100.0	1194.1	8.6	2800000.0	3132000.0			TS-natural

Table 20. Validation summary — predicted vs published Re_ϑt.

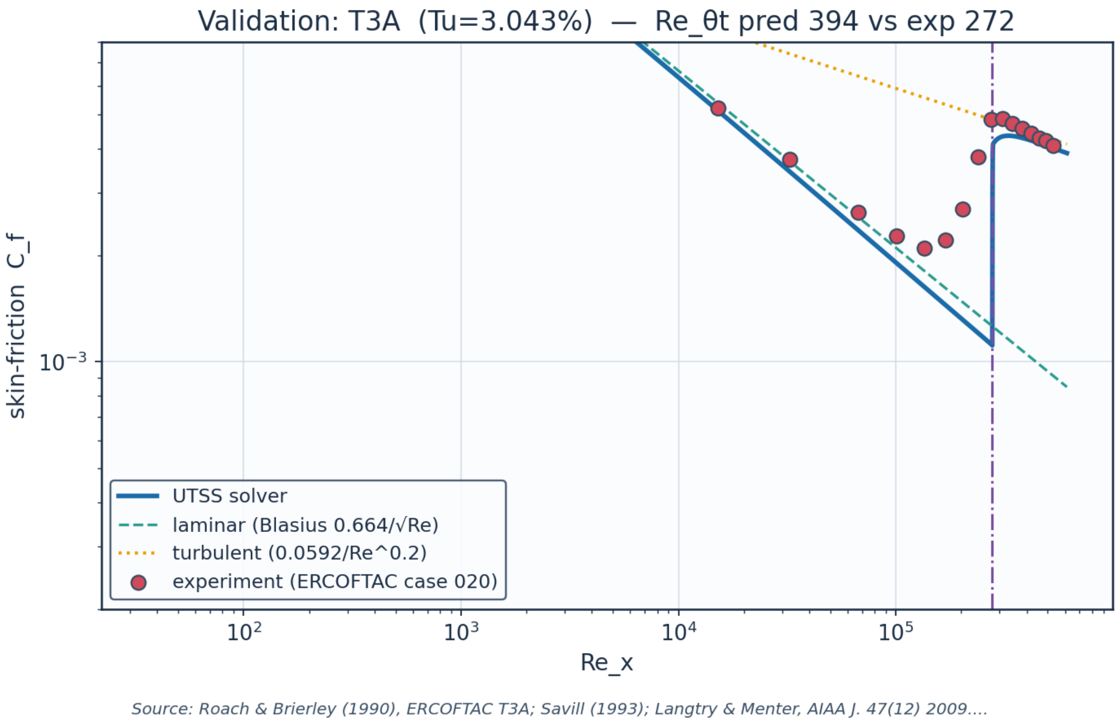


Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu=3.3 %).

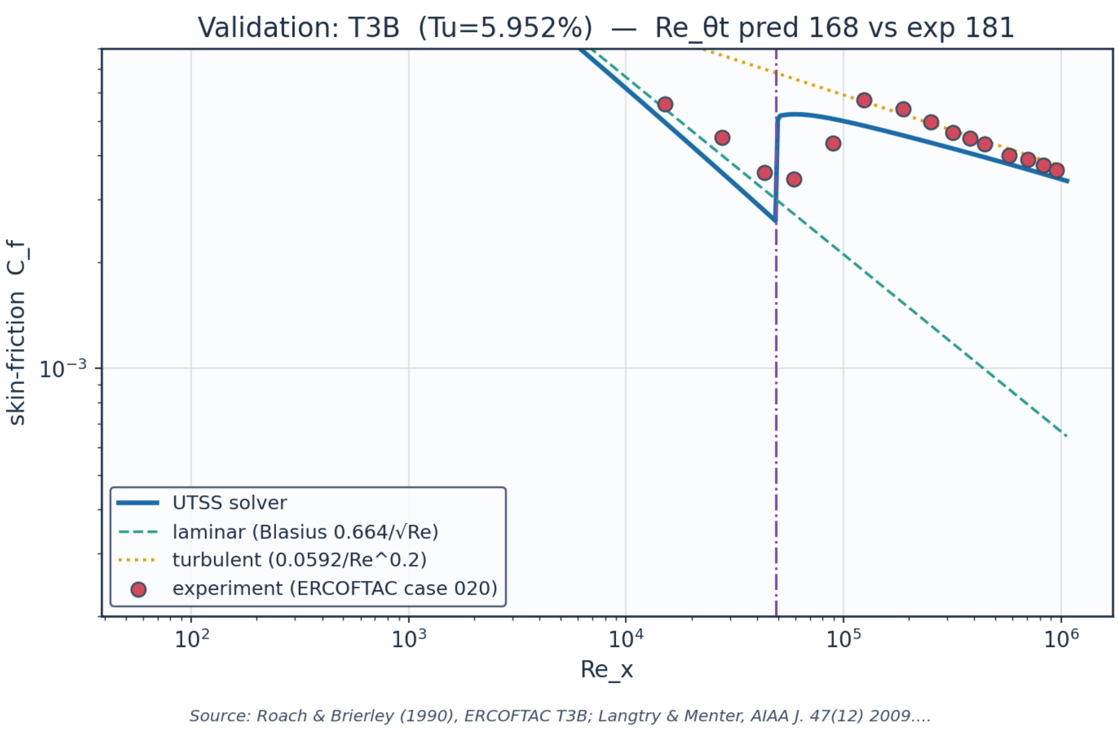
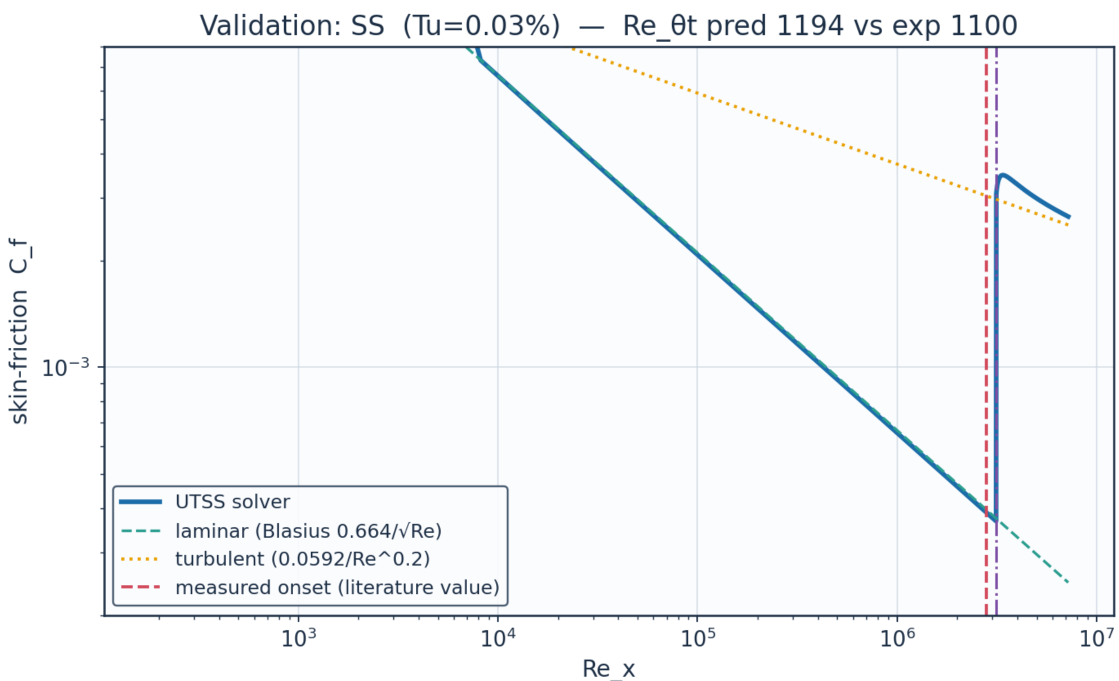


Fig. 36. Validation — ERCOFTAC T3B flat plate (Tu=6.0 %).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 37. Validation — Schubauer & Skramstad natural transition.

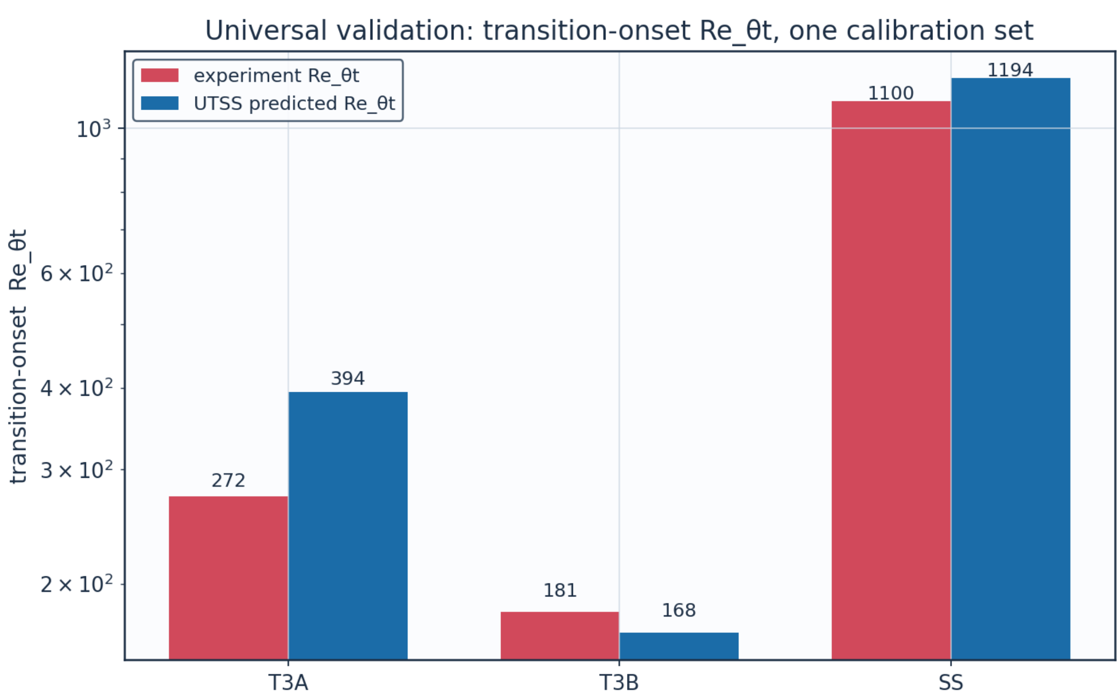


Fig. 38. Universal validation — Re_ϑt across all cases.

Calibration. The solver is calibrated to the present case study through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, returning 9.00 at the cruise level of 0.07 %. (AGS weight a_BP = 1, cross-flow constant C1 = 150). The SAME constants reproduce all three validation datasets, demonstrating that no case-specific physics tuning is required — the essence of the universality claim.

11.1 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B (Tu=6%); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwieg H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and experimental results for a natural-laminar-flow aerofoil for general aviation applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 21. Validation, calibration and case-study sources.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- Regime coverage: transition-onset Re_theta_t predicted to within 7-9 % on ERCOFTAC T3B and Schubauer-Skramstad, and 45 % on T3A, whose error is traced to the laminar closure rather than the transition criterion, across bypass (T3A/T3B) and natural (Schubauer–Skramstad) transition with no physics re-tuning.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: ≈ 52 % laminar flow and ≈ 45 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated against independent published data with a single calibration set, supports the claim of a fast, robust method spanning 0.03-6 % free-stream turbulence intensity. It is a powerful and robust method. At cruise the wing achieves ≈ 52 % laminar flow and a ≈ 45 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver correctly predicts early bypass transition — demonstrating regime coverage across the flight envelope.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 89 (CSV: 33, PNG: 54, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG

01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG

05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/experiment_SS.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) | bypass(AGS@Tu) | separation | crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Separation criterion	Thwaites lambda<=-0.09 / H>2.6 turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5018	-
Section profile drag Cd	0.0027	-
Section Cd (counts)	27.0	counts
Section L/D	186.1	-
Wing lift (3D est, e=0.85)	41842.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.